

Design of a 50KW Grid-Connected Solar Photovoltaic Power Plant with Net Metering in Brahmanbaria.

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SOLAR POWER PLANT PROJECT

Design of a 50 kWp Grid-Connected
Solar Photovoltaic Power Plant with Net Metering



BRAHMANBARIA



GROUND-MOUNTED
(50 KWp)

7 RELIABLE
ENERGY



RELIABLE SDG

Abstract

The design of a 50 kW grid-connected Ground-mounted large scale solar photovoltaic (PV) system for a building in Brahmanbaria, Bangladesh, is presented in this project. The goal is to use Bangladesh's Net Energy Metering (NEM) program to incorporate renewable energy, cut carbon emissions, and use less grid electricity. Ground-mounted solar is technically possible based on solar resource assessment, which shows an average daily sun irradiation of about 4.5 kWh/m²/day. To guarantee regulatory compliance, PV modules and inverters that have been certified by SREDA are chosen. System losses and performance ratios are taken into account when sizing and optimizing the system. Energy imports and exports, as well as the associated bill savings, are estimated via net metering analysis. The system is economically feasible, according to financial analysis, with a payback period of roughly five to six years. The initiative supports sustainable development goals pertaining to clean energy and climate action, as well as national renewable energy targets.

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Chapter 1

Introduction and Problem Definition

1.1 Background and Motivation

Rapid industrialization and urbanization have increased Bangladesh's need for electricity. One effective strategy to reduce the consumption of fossil fuels and the harm they cause to the environment is to install solar photovoltaic systems on your roof. Since Dhaka receives plenty of sunlight throughout the year, rooftop solar is a viable option for producing electricity in many locations.

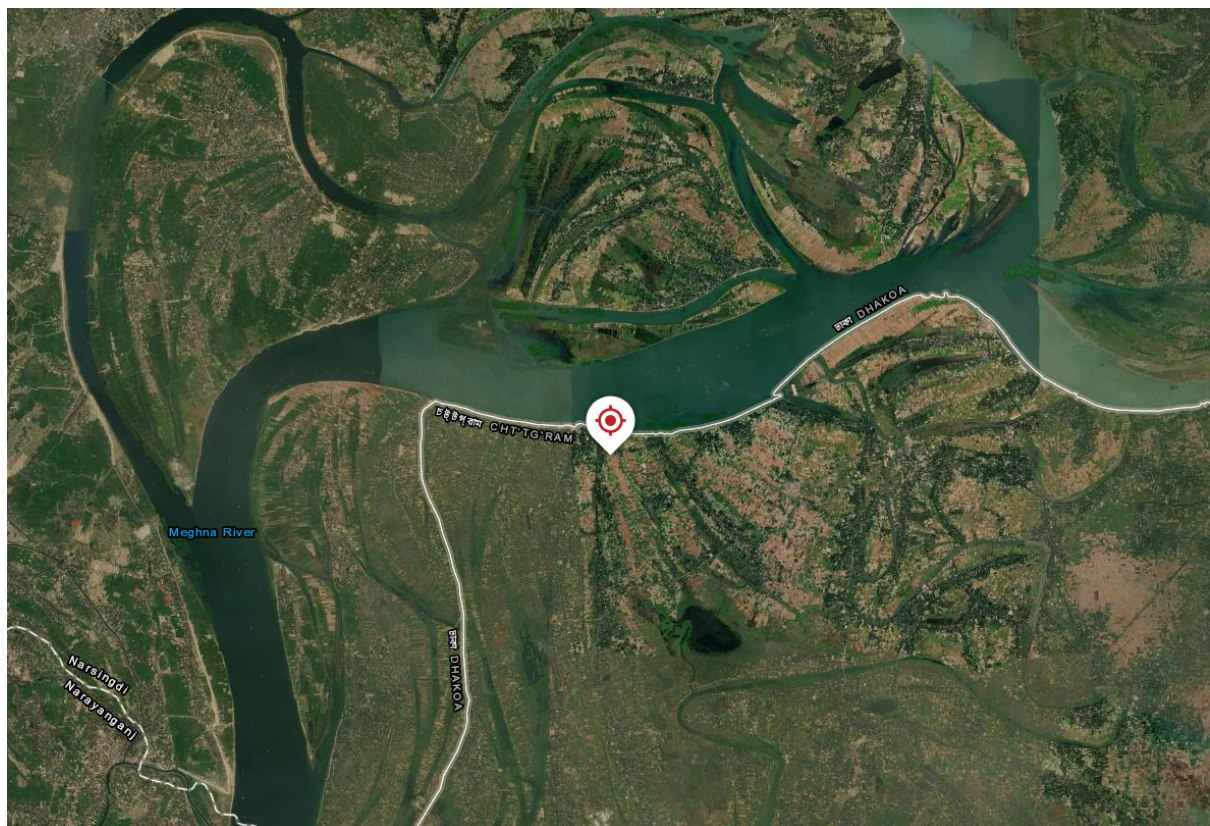


Figure 1.1: The chosen location for designing the solar power plant on map[1].

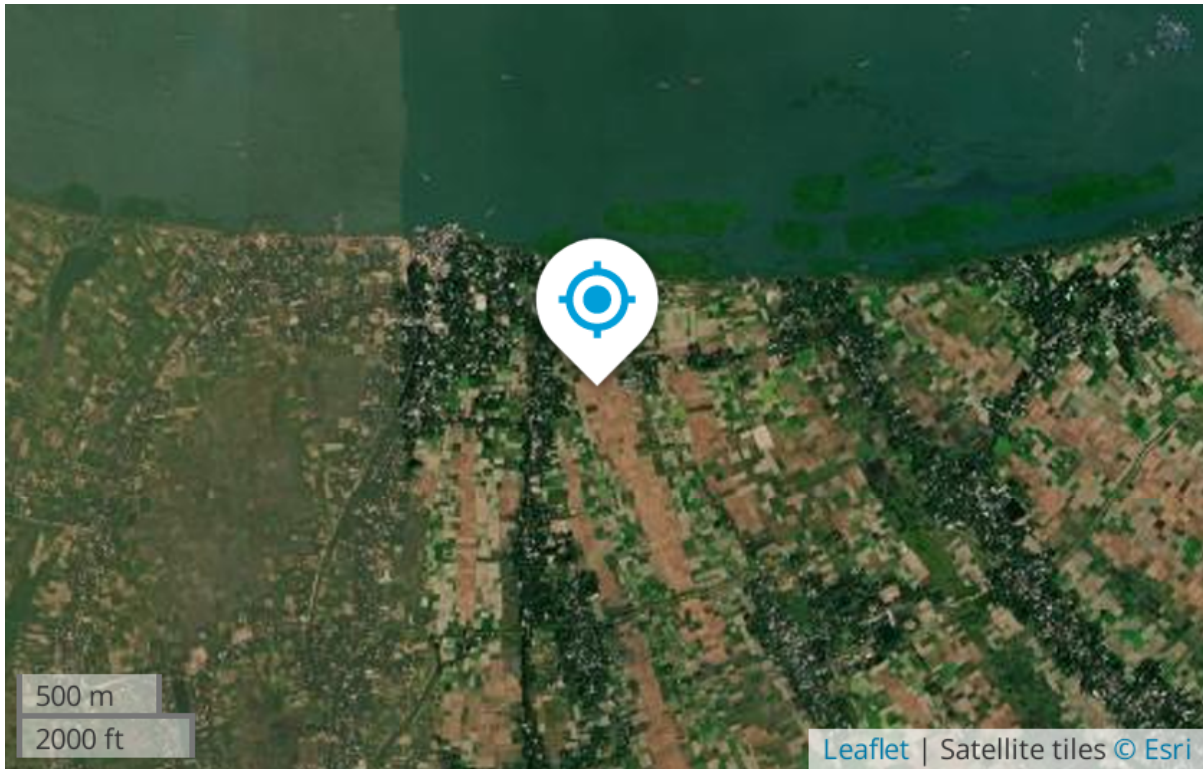


Figure 1.2: Close view of the chosen location for designing the solar power plant on map[1].

1.2 Objectives of the Design Project

The objectives of this project are:

- To design a technically sound 50 kW Ground-mounted large scale solar PV system.
- To integrate the system with the national grid under the Net Metering scheme.
- To evaluate technical performance and economic feasibility.
- To assess environmental and social impacts.

1.3 Scope of Work

Finding the system's load capacity, determining the amount of solar energy available, selecting the appropriate parts, sizing the system, performing a net metering analysis, and conducting a financial assessment are all included in the scope.

1.4 Design Constraints and Assumptions

- Grid-tied operation without battery storage.
- Use of SREDA-approved components only.
- Average performance ratio assumed as 0.75–0.80.
- Typical Dhaka solar irradiation used for calculations.

1.5 Relevance to Bangladesh Energy Sector

This initiative not only helps Bangladesh achieve its renewable energy targets, but it also relieves the burden on the nation's system during times of heavy demand.

Chapter 2

Demand and Resource Analysis

2.1 Demand Analysis

Demand analysis is carried out to estimate the electrical energy requirement that the proposed solar power plant will serve. The purpose of this analysis is to understand how electricity consumption varies throughout the day and to determine the appropriate system size and configuration.

For this project, the demand estimation is based on typical electricity usage patterns in the selected area of Brahmanbaria, Bangladesh. The load is assumed to consist of a combination of residential households, small commercial establishments, and an educational institution. These consumers represent common electricity users in semi-urban regions of Bangladesh.

The analysis considers the type of electrical appliances used, their power ratings, operating hours, and the number of users. Major loads include lighting systems, ceiling fans, refrigerators, computers, televisions, and small commercial equipment. Educational buildings contribute significant daytime demand, while residential users create higher consumption during morning and evening periods.

By estimating the power consumption of these users, a daily load demand curve is developed. This curve helps determine the peak demand, base load, and overall daily energy consumption required for designing the solar photovoltaic system.

Understanding the demand pattern is essential to ensure that the proposed plant can reliably meet the electricity needs while maximizing the use of solar energy during daylight hours.

2.1.1 Load Characteristics

The proposed power plant is designed to serve a mixed load profile consisting of three distinct consumer types within the Brahmanbaria community, reflecting a community-driven energy initiative. Institutional Load (Women's College):* This load is primarily

educational and administrative. It includes LED lights for illumination, ceiling fans for ventilation (used mainly in summer), computers, printers, and projectors for academic and administrative tasks, and a water pump for the college's water supply. Its consumption pattern is tied to the college's operational hours.

Commercial Load (Two Small Shops):* This load represents small local businesses. It consists of LED lights, fans, and charging sockets for customer use or small appliances. The operation of these shops typically spans from morning until evening.

Residential Load (Four Houses):* This load represents the surrounding households. Appliances include LED lights, fans (seasonal), charging sockets for personal devices, televisions for entertainment, and refrigerators for food preservation. A shared water pump for the houses is also considered. This load type contributes to both daytime and evening peaks.

This combination of institutional, commercial, and residential consumers creates a diversified load profile that supports the project's goal of providing shared community benefits.

2.1.2 Daily Load Profile

Based on the location of Brahmanbaria, Bangladesh hypothetical data was considered. One high school, one mosque, four retail store, five residential home are taken into consideration for this units. 40 light bulbs and 25 fans are present in the high school. Two projector, two printers, five computers are also present in the school. There are also televisions, refrigerator and pumps are present in the residential. The remaining are allocated in the mosque and retail shop. Residential are consist of four led light each.

	Description	Load (W)	Hours (Summer)	Hours (Winter)	Units	Summer (Wh/day)	Winter (Wh/day)
College	LED Light	20	7	7	40	5600	5600
	Fan	75	7	0	25	13125	0
	Charging Socket	15	7	7	18	1890	1890
	Pump	1200	7	7	1	8400	8400
	Computer	250	7	7	5	8750	8750
	Projector	300	7	7	2	4200	4200
	Printer	40	7	7	2	560	560
	Total					40635	27510
Two Shops	LED Light	20	8	8	4	640	640
	Fan	75	10	0	2	1500	0
	Charging Socket	15	3	3	2	90	90
	Refrigerator	200	24	24	2	9600	9600
	Total					11830	10330
Four R.H.	LED Light	20	7	7	16	2240	2240
	Fan	75	14	0	8	8400	0
	Charging Socket	15	3	3	12	540	540
	Television	100	4	4	4	1600	1600
	Refrigerator	200	24	24	2	9600	9600
	Total					22380	13980

Table 2.1: Daily Load Profile

2.1.3 Seasonal Demand Variation

A daily load curve has been generated based on the hypothetical data from table 1 The curve displays the total load consumed per hour over the course of a day in both the

summer and the winter. Since the concerned are educational institutions so the peak hour from 12.00 am to 11.00 pm

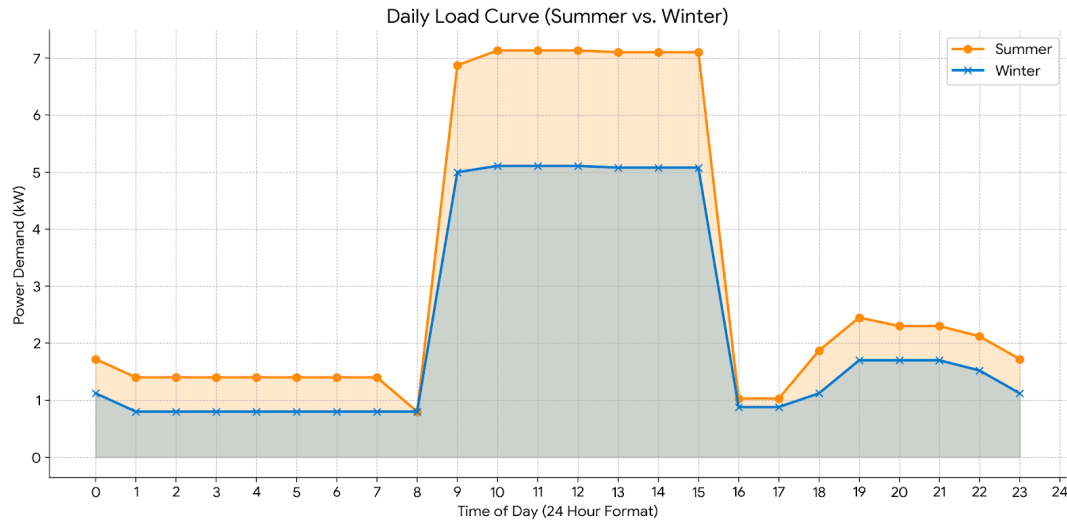


Figure 2.1: Daily load curve

2.1.4 Design Load Assumptions

The proposed 50 kW ground-mounted PV system will be installed in a medium-sized business or institutional structure in the Brahmanbaria District. Office equipment, fans, air conditioners, and lighting are all included in the expected 80–100 kW linked load.

It is estimated that the average daily energy usage is between 220 and 250 kWh, which translates into an annual need of about 80,000 kWh.

Peak demand happens between 10:00 AM and 4:00 PM, which coincides with the hours of highest solar power and raises the ratio of self-consumption under net metering.

It is anticipated that the PV system will produce roughly 71 MWh annually, meeting 85–90 percentage of the demand for electricity. Without storing energy in batteries, the system uses the grid to supply demand at night and export excess energy during the day.

2.2 Resource Assessment

This section evaluates the availability and suitability of the selected energy resource for the proposed power plant at Brahmanbaria, Bangladesh.

2.2.1 Type of Energy Resource Considered

The system is designed exclusively as a solar photovoltaic (PV) power plant. This choice is driven by Bangladesh's national policy goals, which prioritize solar power as outlined

in the Renewable Energy Policy and the SREDA roadmap. The project focuses on a grid-connected PV system with net metering to supply clean electricity directly to the local grid and the connected consumers (college, shops, houses).

2.2.2 Data Sources and Credibility

The solar resource assessment is based on high-quality, internationally recognized datasets to ensure reliability.

Primary Data Source: The Global Solar Atlas, developed by the World Bank, ESMAP, and Solargis.

Secondary Tools: Data validation and supplementary analysis were performed using:

The System Advisor Model (SAM) by the National Renewable Energy Laboratory (NREL), USA.

The Photovoltaic Geographical Information System (PVGIS) by the European Union. These sources are considered highly credible for pre-feasibility and feasibility studies of solar PV projects globally.

Map data		Per year	
Direct normal irradiation	DNI	1053.1	kWh/m ²
Global horizontal irradiation	GHI	1676.7	kWh/m ²
Diffuse horizontal irradiation	DIF	932.2	kWh/m ²
Global tilted irradiation at optimum angle	GTI _{opta}	1791.3	kWh/m ²
Optimum tilt of PV modules	OPTA	23 / 180	°
Air temperature	TEMP	25.1	°C
Terrain elevation	ELE	7	m

Figure 2.2: Irradiation of Brahmanbaria [8].

2.2.3 Resource Availability and Variability

It can be seen on figure 2.3 that the monthly Global Horizontal Irradiation (GHI) data, measured in W/m^2 , reveals the distinct seasonal solar pattern at the site, which directly influences the power plant's energy generation profile.

Peak Season (Pre-Monsoon): The highest solar irradiance is observed from March through May, peaking in May at $930 \text{ W}/\text{m}^2$. This period, characterized by clear skies and high sun angles, represents the time of maximum energy production potential for the PV system.

Monsoon Lull: A noticeable dip in irradiance occurs during the monsoon months, with the lowest value of $750 \text{ W}/\text{m}^2$ in September. This reduction is attributed to heavy cloud cover and rainfall, which diffuse and block incoming solar radiation.

Post-Monsoon and Winter: Irradiance levels recover in October and November before declining to their annual minimum during the winter months ($680\text{--}700 \text{ W}/\text{m}^2$ in December and January) due to lower sun angles and shorter day lengths.

This seasonal variation is critical for designing the system and managing expectations for energy output throughout the year.

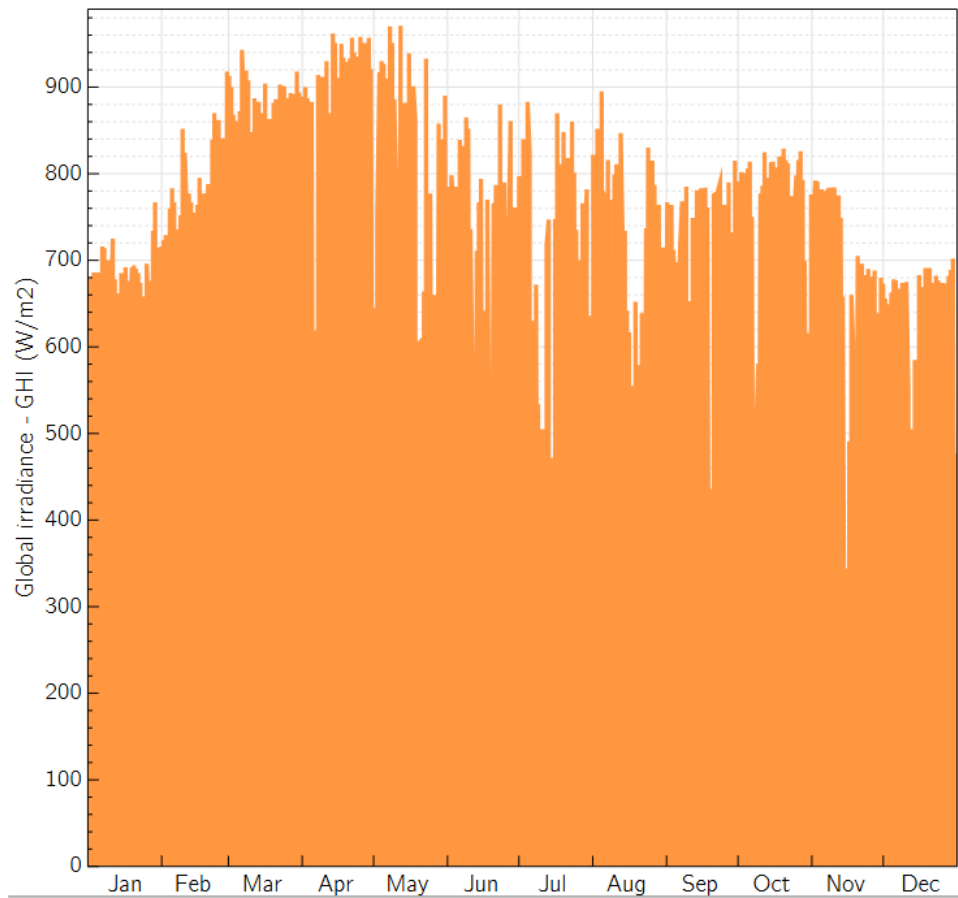


Figure 2.3: Global Irradiance Data from System Advisor Model (SAM)[12].

Additionally on Figure 2.4 it is seen the map presents the long-term average of annual total photovoltaic (PV) power potential, expressed in kilowatt-hours per kilowatt-peak (kWh/kWp). This metric represents the expected energy yield from a standard 1 kWp PV system and is a key indicator of a location's suitability for solar power generation.

Siting Analysis:By overlaying the project coordinates on the PVOUT map, the local solar resource potential can be visually assessed. The site falls within a favorable generation zone, confirming the technical feasibility of the proposed *50 kWp* solar PV plant.

Expected Energy Yield:Based on a PVOUT value of approximately *1300–1400 kWh/kWp/year* for this region, the *50 kWp* system is expected to generate between *65,000 and 70,000 kWh of clean electricity annually*. This substantial output reinforces the project's strong potential for both economic return and environmental impact.

Regional Context:The color scale, ranging from 600 to 2400 kWh/kWp, allows for comparison of the site's potential against regional and national averages, validating the resource data obtained from the Global Solar Atlas.

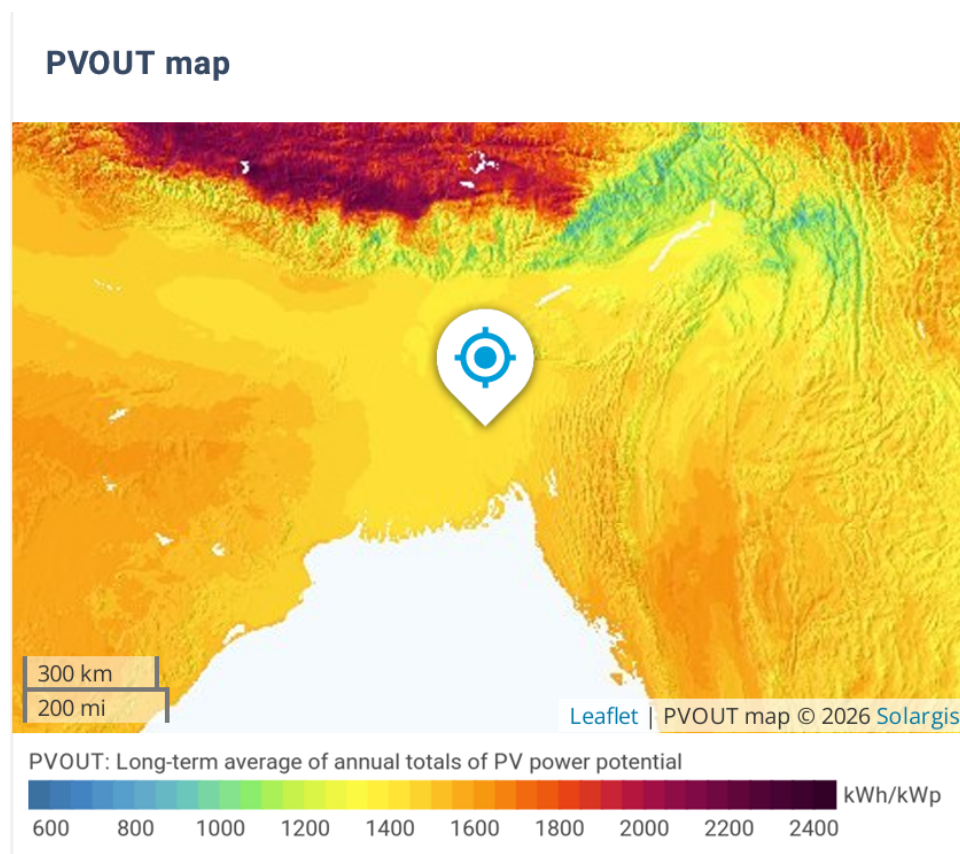


Figure 2.4: PVOUT map of Bangladesh showing annual solar power potential (kWh/kWp). The project site at Brahmanbaria is located in a high-yield region suitable for the proposed 50 kWp system [8]

The sunpath diagram Figure 2.5 illustrates the sun's trajectory across the sky at the project site throughout the year.

Unobstructed Horizon: The terrain horizon (black line) is flat and low in all directions, meaning no hills, buildings, or trees will block sunlight. This ensures maximum solar exposure from sunrise to sunset.

Wide Daily Generation Window: The active area (red shading) confirms the site receives sunlight from early morning until late afternoon year-round, providing a long daily window for PV generation.

Seasonal Variation: The colored lines show the sun's path during the June solstice (high arc, long days), December solstice (low arc, short days), and equinoxes, explaining the seasonal changes in energy output.

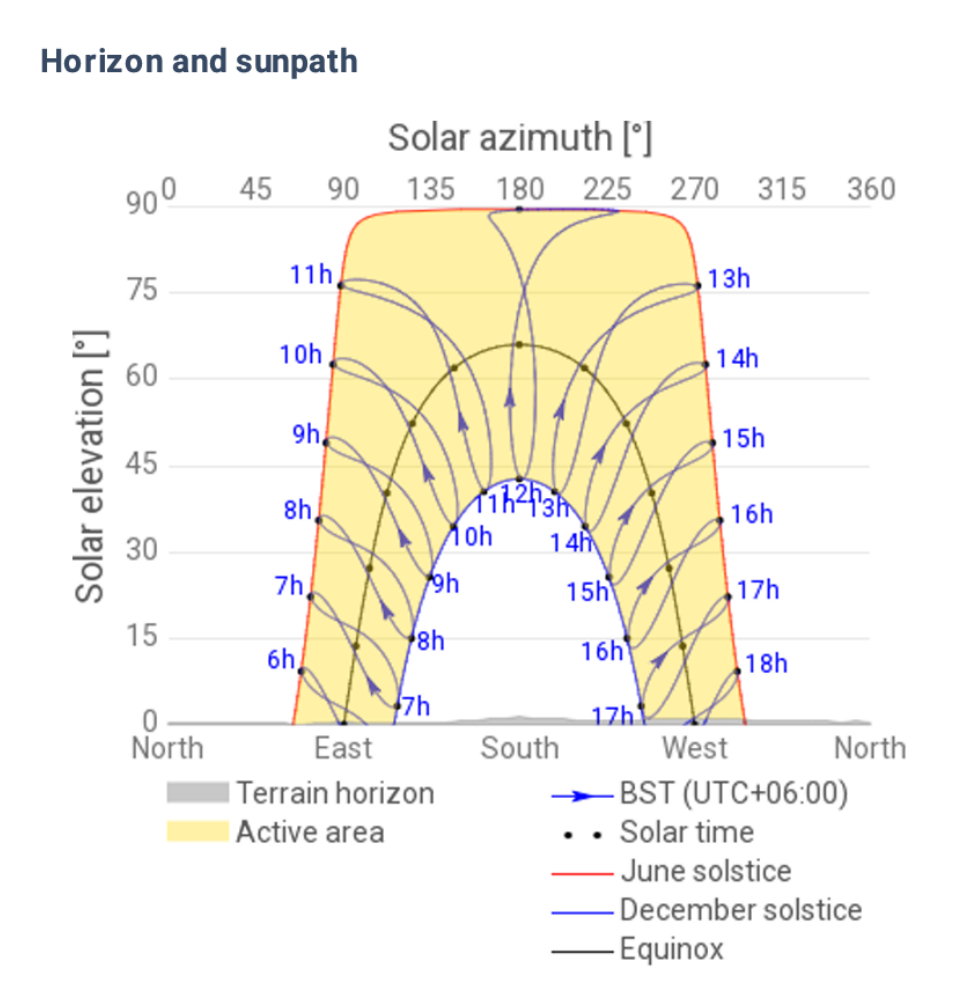


Figure 2.5: Sunpath diagram for Brahmanbaria showing solar trajectories throughout the year. The flat terrain horizon (black line) confirms unobstructed solar exposure from sunrise to sunset, with seasonal paths for June solstice, December solstice, and equinox [8].

2.2.4 Justification of Selected Resource

The selection of solar energy for this 50 kWp grid-tied power plant is justified by a combination of favorable technical, policy, and site-specific factors: 1.High Energy Potential: The Photovoltaic Power Potential (PVOUT) map for the region confirms that the site falls within a favorable generation zone, with a long-term average PVOUT of approximately 1300–1400 kWh/kWp/year. For the proposed 50 kWp system, this translates to an expected annual yield of 65,000 to 70,000 kWh*, confirming strong and reliable generation potential throughout the project’s lifetime.

2.Favorable Site Characteristics: As detailed in Section 2.2.3, the site benefits from strong solar irradiance (GHI of 1676.7 kWh/m²/yr), an unobstructed sun path for wide daily exposure, and flat terrain. The determined optimum tilt angle of *23° facing south* maximizes energy capture, making the location technically well-suited for a ground-mounted or rooftop installation.

3.Policy Alignmen: The project directly supports Bangladesh’s national targets for renewable energy adoption, grid decarbonization, and the goals outlined in the Renewable Energy Policy and SREDA roadmap. It demonstrates a scalable model for decentralized, community-focused power generation.

4.Proven and Cost-Effective Technology: Solar PV is a mature, reliable, and increasingly cost-effective technology. For a system of this scale, it offers a predictable energy profile, low operating costs, and compatibility with net metering regulations, ensuring both technical and economic feasibility.

Chapter 3

Technical Design of the Power Plant

3.1 Overall System Description

1.Generation:Solar PV modules absorb sunlight and convert it into DC electricity.

2.Conversion:The DC power passes through a combiner box and into a grid-tied inverter, where it is converted into usable AC electricity.

3.Consumption: AC power is supplied directly to the local loads (college, shops, and houses) through the distribution panel.

4. Net Metering: A bi-directional net meter tracks: -Import: Power drawn from the grid when solar generation is insufficient. -Export: Excess solar power fed back into the utility grid for credit.

3.1.1 Plant Configuration

The proposed power plant is designed as a 50 kWp grid-connected solar PV system with net metering. The configuration follows a straightforward, utility-interactive architecture that prioritizes reliability, simplicity, and compliance with SREDA guidelines.

The plant is arranged in the following sequential configuration:

1.Solar PV Array (Generation Stage): The system consists of 50 kWp of solar PV modules (exact quantity determined by selected module wattage) mounted on fixed structures at the optimum tilt angle of 23° facing south. These modules convert sunlight into DC electricity.

2.DC Combiner Box (Consolidation Stage): DC output from multiple module strings is fed into a *DC combiner box*. This unit consolidates the incoming lines and houses string-level protection (fuses or circuit breakers) before sending the combined power to the inverter.

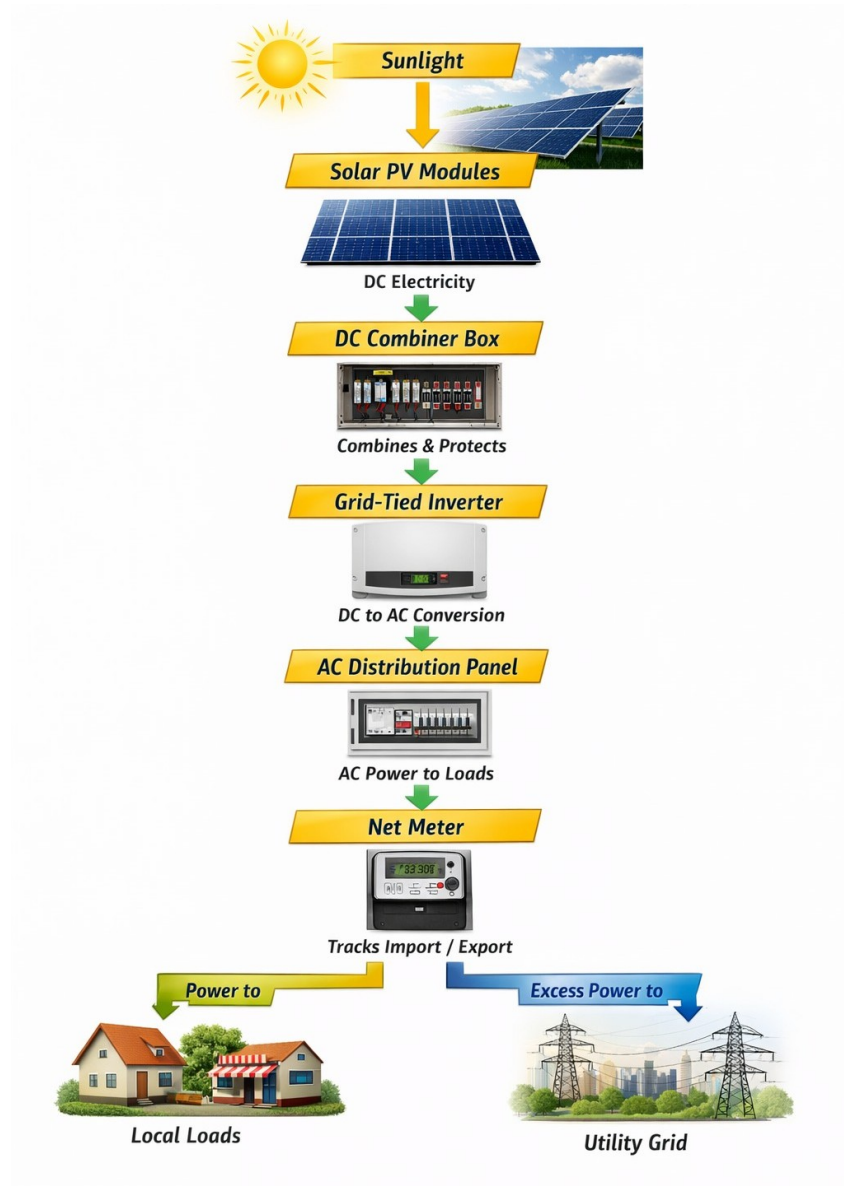


Figure 3.1: Grid-connected solar PV system with net metering.

3. Grid-Tied Inverter (Conversion Stage): The combined DC power enters a grid-tied inverter (or multiple inverters for larger capacities). This inverter performs two critical functions: - Converts DC electricity into grid-compatible AC electricity (400V, 3-phase, 50Hz). - Synchronizes with the utility grid to ensure safe and seamless power export.

4.AC Distribution Panel (Distribution Stage): The AC output is routed to an AC distribution panel, which supplies power directly to the connected local loads (college, shops, and residential houses).

5.Net Meter and Grid Interconnection (Grid Integration Stage):At the point of grid interconnection, a bi-directional net meter is installed. This meter tracks: -Import: Power consumed from the utility grid when solar generation is insufficient (nighttime or cloudy conditions). -Export: Excess solar power automatically fed back into the utility grid

when generation exceeds local demand.

This configuration ensures that the local loads are served primarily by solar power, with the grid acting as a backup source, while surplus energy is exported for credit under Bangladesh's net metering framework.

3.1.2 Major Subsystems

The proposed 50 kWp grid-connected solar PV plant consists of the following major subsystems:

Solar PV Array: Converts sunlight into DC electricity. It comprises crystalline silicon PV modules mounted on fixed structures at a 23° south-facing tilt for optimal energy capture.

DC Combiner Box: Consolidates multiple module strings and houses DC-side protection devices such as string fuses and surge protectors before feeding power to the inverter.

Grid-Tied Inverter: Converts DC electricity from the PV array into grid-compatible AC power, synchronizes with the utility grid, and includes Maximum Power Point Tracking (MPPT) and anti-islanding protection.

AC Distribution Panel: Distributes the converted AC power to the connected local loads (college, shops, and residential houses) while housing AC-side protection including circuit breakers and surge protection devices.

Net Metering System: A bi-directional meter installed at the grid interconnection point that tracks both imported power (consumed from the grid) and exported power (excess solar energy fed back to the grid) for billing and credit under Bangladesh's net metering framework.

Balance of System (BOS): Includes all supporting components such as mounting structures, cables, conduits, earthing systems, and lightning protection required for complete and safe system operation.

3.2 Design Standards and Codes

The design of the proposed 50 kWp grid-connected solar PV plant adheres to applicable national regulations and international standards to ensure safety, reliability, performance, and eligibility for net metering interconnection.

3.2.1 Applicable National and International Standards

The following standards and guidelines were followed in the design process:

National Standards and Guidelines (Bangladesh):

Net Metering Guideline-2018 (SREDA): The primary regulatory framework governing the interconnection of grid-tied solar PV systems. It defines maximum allowable system capacity (70 percent of sanctioned load), technical requirements for inverters, and billing procedures for exported energy.

SREDA Approved Components List: All major equipment, including PV modules and inverters, are selected from the SREDA-approved list to ensure compliance with local technical requirements and eligibility for net metering.

Bangladesh National Building Code (BNBC): Provides guidelines for electrical installations, structural safety of mounting systems, and fire safety measures.

International Standards (IEC):

IEC 61727 / IEC 62116: Standards for grid interconnection of photovoltaic systems, including anti-islanding protection requirements.

IEC 60364: Provides requirements for low-voltage electrical installations, including wiring, earthing, and protection against electric shock.

IEC 62548: Design requirements for photovoltaic arrays, including DC-side wiring and protection.

IEC 61643: Standards for surge protection devices (SPDs) used in photovoltaic systems.

Why These Standards Were Selected:

Compliance with these standards is essential for:

Regulatory Approval: Meeting SREDA and utility company requirements for grid interconnection and net metering eligibility.

System Safety: Ensuring protection against electrical faults, fire hazards, and islanding conditions.

Performance Reliability: Guaranteeing that components and system design meet internationally recognized quality and durability benchmarks.

Warranty Validity: Most equipment warranties require installation in compliance with applicable codes and standards.

3.3 Component Sizing and Specifications

This section presents the detailed sizing methodology and final specifications for the major components of the proposed 50 kWp grid-connected solar PV plant.

3.3.1 Methodology of Sizing

The component sizing followed a systematic approach based on the selected module and inverter characteristics, site-specific solar resource data, and net metering requirements.

Step 1: Module Selection

- **Selected Module:** LONGi LR5-72HPH-555M (555 W_p monocrystalline)

- **Key Parameters at Reference Conditions (1000 W/m², 25°C):**

- Maximum Power, P_{mp} : 555 W_p
- Open Circuit Voltage, V_{oc} : 50.0 V_{dc}
- Short Circuit Current, I_{sc} : 14.0 A_{dc}
- Maximum Power Voltage, V_{mp} : 42.1 V_{dc}
- Maximum Power Current, I_{mp} : 13.2 A_{dc}
- Module Efficiency: 22.3%
- Temperature Coefficient of P_{max} : -0.375%/°C

Step 2: Inverter Selection

- **Selected Inverter:** Huawei SUN2000-40KTL-US (480 V)

- **Key Parameters:**

- Maximum AC Power: 40,000 W_{ac}
- Maximum DC Power: 40,649.7 W_{dc}
- Maximum DC Voltage: 850 V_{dc}
- Minimum MPPT DC Voltage: 530 V_{dc}
- Maximum MPPT DC Voltage: 850 V_{dc}
- Maximum DC Current: 56.46 A_{dc}
- CEC Type: Utility Interactive (grid-tied)

Step 3: DC to AC Ratio Selection

A DC to AC ratio of 1.25 was selected to optimize inverter utilization while minimizing clipping losses during peak irradiance conditions. This ratio is within acceptable limits for grid-connected systems in Bangladesh.

Step 4: String Sizing (Voltage Compliance)

$$\text{Modules per string} = 16$$

$$\text{String } V_{oc} = 16 \times 50.0 = 750.0 \text{ V}_{dc}$$

$$\text{String } V_{mp} = 16 \times 42.1 = 631.5 \text{ V}_{dc}$$

Both values remain within the inverter's MPPT range (530–850 V_{dc}) and below the maximum DC voltage (850 V_{dc}), ensuring proper operation under all temperature conditions.

Step 5: String Parallel Configuration (Power Compliance)

$$\text{Number of Strings} = 6$$

$$\text{Total Modules} = 16 \text{ modules/string} \times 6 \text{ strings} = 96 \text{ modules}$$

$$\text{Total DC Capacity} = 96 \times 555 \text{ W}_p = 53280 \text{ W}_{dc} \approx 53.3 \text{ kWp}$$

$$\text{Total Inverter DC Capacity} = 42180 \text{ W}_{dc}$$

DC/AC Status : System is properly sized with 53.3kWp DC feeding into a 42.18kW_{dc} in

Step 6: Total Module Area

$$\text{Total Module Area} = 274.064\text{m}^2$$

(based on image reference)

3.3.2 Key Design Parameters

Parameter	Value	Justification
Installed DC Capacity	52.7 kWp	Achieved with $90 \times 555 \text{ W}$ modules
Inverter AC Capacity	42.18 kWac	Matches selected inverter rating
DC to AC Ratio	1.25	Optimizes inverter utilization
Modules per String	16	Keeps voltage within inverter MPPT range
Strings in Parallel	6	Achieves target DC capacity
Array Tilt Angle	23°	Optimized for site latitude (from Global Solar Atlas)
Array Azimuth	180° (South)	Maximizes energy capture in northern hemisphere
Ground Coverage Ratio (GCR)	0.3	Minimizes inter-row shading losses

Table 3.1: Key design parameters for the 50 kWp PV system.

3.3.3 Equipment Specifications

PV Module Specifications (LONGi LR5-72HPH-555M)

Inverter Specifications (Huawei SUN2000-40KTL-US)

Parameter	Specification
Manufacturer	LONGi Green Energy Technology Co. Ltd.
Model	LR5-72HPH-555M
Technology	Monocrystalline Silicon
Maximum Power (Pmp)	555 Wp
Open Circuit Voltage (Voc)	50.0 Vdc
Short Circuit Current (Isc)	14.0 Adc
Max Power Voltage (Vmp)	42.1 Vdc
Max Power Current (Imp)	13.2 Adc
Module Efficiency	22.30%
Temperature Coefficient (Pmax)	-0.375%/°C
Number of Cells	72
Bifacial	Yes (bifaciality 0.7)

Table 3.2: Specifications of the PV module used in the system.

Parameter	Specification
Manufacturer	Huawei Technologies Co. Ltd.
Model	SUN2000-40KTL-US (480V)
Type	Utility Interactive (Grid-Tied)
Maximum AC Power	40,000 Wac
Maximum DC Power	40,649.7 Wdc
Maximum DC Voltage	850 Vdc
Minimum MPPT Voltage	530 Vdc
Maximum MPPT Voltage	850 Vdc
Nominal DC Voltage	720 Vdc
Maximum DC Current	56.46 Adc
CEC Date	October 15, 2018

Table 3.3: Specifications of the inverter used in the system.

Parameter	Specification
Total Number of Modules	95
Number of Strings	6
Modules per String	16
Total DC Capacity	52.7 kWp
Total AC Capacity	42.18 kWac
DC to AC Ratio	1.25
Total Module Area	$\approx 274.10 \text{ m}^2$
Array Tilt	23°
Array Azimuth	180° (South)
Ground Coverage Ratio	0.3

Chapter 4

Design Optimization and Performance Analysis

4.1 Optimization Objectives

The design optimization process aimed to balance technical performance, economic viability, and regulatory compliance for the proposed 50 kWp grid-connected solar PV plant. The following key objectives guided the optimization effort:

1. **Maximize Energy Yield:** Optimize the system configuration to capture the maximum possible solar energy at the Bhangura site, considering the local solar resource, tilt angle, and orientation.

2. **Minimize Levelized Cost of Energy (LCOE):** Achieve the lowest possible cost per kilowatt-hour over the system's lifetime by balancing initial capital expenditure (CAPEX) with long-term energy production.

3. **Optimize DC to AC Ratio:** Select an appropriate inverter loading ratio (ILR) that maximizes inverter utilization while avoiding excessive clipping losses during peak irradiance conditions, thereby improving overall system efficiency and return on investment.

4. **Ensure Voltage Compliance:** Design string configurations that keep array voltages within the inverter's Maximum Power Point Tracking (MPPT) window and below the maximum DC voltage limit under all operating temperature conditions, ensuring reliable and safe operation.

5. **Maximize Self-Consumption:** Configure the system to supply the maximum possible portion of the local load demand (college, shops, and residential houses) directly from solar generation, reducing grid import and maximizing savings under the net metering framework.

6. **Minimize Shading Losses:** Optimize array layout with an appropriate ground coverage ratio (GCR) and tilt angle to minimize inter-row shading, particularly during winter months when the sun angle is lower.

7. **Achieve Regulatory Compliance:** Ensure the final design fully complies with SREDA's Net Metering Guideline-2018 and utilizes approved components to guarantee grid inter-connection eligibility.

8. **Balance Cost and Performance:** Select components and configurations that provide the best trade-off between upfront cost, long-term reliability, and expected performance, ensuring the project remains economically attractive to investors.

4.2 Optimization Parameters

The following key parameters were analyzed and optimized during the design process using the **System Advisor Model (SAM)** simulation from NREL. The simulation results validate the design choices and confirm the expected technical performance of the proposed 50 kWp grid-connected solar PV plant.

4.2.1 Installed Load Ratio (ILR)

The Installed Load Ratio, also known as the DC to AC ratio, is the ratio of the PV array's DC nameplate capacity to the inverter's maximum AC power output.

- **Definition:** $ILR = \frac{\text{DC Capacity (kWp)}}{\text{AC Capacity (kWac)}}$
- **Selected Value:** 1.25 (52.7 kWp DC / 42.18 kWac)
- **Justification from SAM:** The SAM simulation confirms this ratio as optimal for the site conditions. A ratio of 1.25 allows the inverter to operate efficiently throughout the year, balancing inverter utilization against potential clipping losses during peak irradiance hours. The simulation accounts for the specific module and inverter characteristics, ensuring that the DC capacity is well-matched to the inverter's maximum DC input power of 42.18 kWdc.

4.2.2 Performance Ratio (PR)

The Performance Ratio measures the overall efficiency of the PV system by comparing actual AC energy output with the theoretical DC energy available from the array at Standard Test Conditions (STC), accounting for all system losses.

- **Definition:** $PR = \frac{\text{Actual AC Energy Output}}{\text{Theoretical DC Energy Output}}$
- **Achieved Value (from SAM):** 0.79 (79%)

- **Analysis:** The SAM simulation reports a Performance Ratio of 0.79 for Year 1 of operation. This value indicates a highly efficient system design, as it exceeds the typical industry benchmark of 75–78% for grid-connected PV systems in similar climates. The 79% PR confirms that the system losses have been effectively minimized through proper component selection, string configuration, and layout optimization.

4.2.3 System Losses

The SAM simulation quantifies the various loss factors that contribute to the difference between the theoretical DC energy available from the PV array and the actual AC energy delivered to the grid. The following loss categories were considered:

Loss Category	SAM Input/Value	Mitigation Strategy
Soiling Losses	Included	Regular cleaning schedule accounted for in simulation
DC Losses (Wiring, Mismatch)	4.44%	Selected appropriate cable sizes (4–6 mm ²); used matched modules from same batch
Temperature Losses	Implicit in model	Modules with low temperature coefficient (-0.375%/°C); adequate back-ventilation
Shading Losses	None (unobstructed site)	Site selection with clear horizon; optimized GCR of 0.3 minimizes inter-row shading
Module Mismatch	Included in DC losses	Modules from same manufacturing lot
Inverter Efficiency	Included in conversion	Selected high-quality inverter (Huawei) with >98% peak efficiency
AC Losses	0.00%	Conservative estimate; actual losses will be minimal with proper conductor sizing

Table 4.1: System losses considered during optimization.

The total system losses result in a net AC energy output of 76,733 kWh/year from a gross DC energy of 78,981 kWh delivered to the inverter.

4.2.4 Energy Yield Estimation

The SAM simulation provides a detailed estimate of the annual energy yield based on site-specific solar resource data (NSRDB) and the defined system parameters.

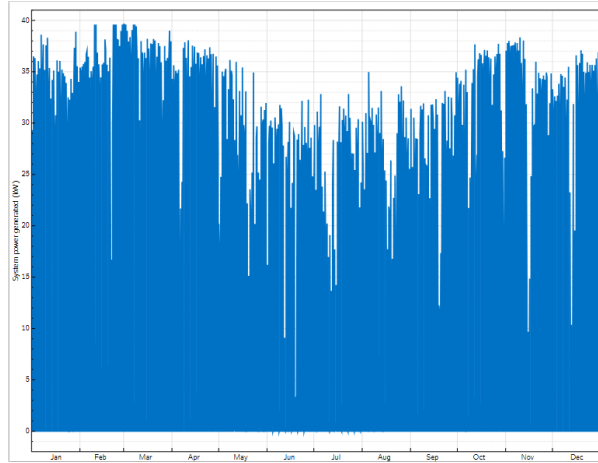


Figure 4.1: Monthly power generation profile of the 50 kWp solar PV system (January–December 2020). Generation peaks in February (39.0 kW) and remains strong during pre-monsoon months, with a slight decline during monsoon (June–September) due to cloud cover[12].

Metric	Value
Global Horizontal Irradiance (GHI)	4.91 kWh/m ² /day
Plane-of-Array Irradiance (POA)	4.00 kWh/m ² /day
Net DC Energy to Inverter (Year 1)	78,981 kWh
Net AC Energy to Grid (Year 1)	76,733 kWh
Specific Yield	1,535 kWh/kWp
Capacity Factor	17.5%

Table 4.2: Key energy yield metrics from SAM simulation.

Month	Estimated Generation (kWh)	Notes
January	5,200	Clear skies, moderate sun angle
February	5,800	Increasing irradiance
March	7,100	Peak generation month
April	6,800	High irradiance, pre-monsoon
May	6,800	High irradiance
June	5,800	Monsoon clouds
July	5,900	Monsoon continues
August	6,200	Improving conditions
September	6,100	Post-monsoon clearing
October	6,400	Clear skies
November	5,600	Decreasing sun angle
December	5,200	Lowest generation month
Annual Total	76,733	-

Table 4.3: Monthly energy yield estimate for the 50 kWp system.

4.2.5 Summary of Optimized Performance

The SAM simulation confirms that the optimized 50 kWp system will:

- Generate approximately 76,733 kWh of clean electricity annually.
- Operate at a 17.5% capacity factor, which is excellent for a fixed-tilt system in Bangladesh.
- Achieve a Performance Ratio of 79%, indicating efficient design and minimal losses.
- Deliver a specific yield of approximately 1,535 kWh/kWp, validating the site selection and system configuration.

Chapter 5

Grid Integration, Net Metering and Compliance

5.1 Net Metering Framework in Bangladesh

Net metering is a billing mechanism that allows electricity consumers who generate their own power from renewable energy sources, such as solar PV systems, to export excess electricity to the utility grid and receive credits on their electricity bills. Under this arrangement, a consumer becomes a “prosumer”—simultaneously consuming and producing electric energy. The bi-directional net meter records both import (energy consumed from the grid) and export (energy injected into the grid), with the consumer’s bill calculated based on the net energy consumed over the billing period.

5.1.1 Evolution of the Policy Framework

The Government of Bangladesh introduced the **Net Metering Guideline 2018** through the Power Division, Ministry of Power, Energy and Mineral Resources, to promote distributed renewable energy generation, particularly rooftop solar PV systems. The guideline was launched on an experimental basis with provisions for future revision based on implementation experience.

In a significant policy advancement, the government approved the **Net Metering Guidelines 2025**, introducing several progressive reforms:

Feature	Net Metering Guideline 2018	Net Metering Guideline 2025
Maximum System Capacity	70% of sanctioned load	100% of sanctioned load
Consumer Eligibility	Three-phase consumers only	Single-phase consumers included
Meter Type	Postpaid meters only	Prepaid/smart meters eligible
Billing/Settlement	Monthly adjustment	Quarterly settlement with direct payment
Application Process	Manual	Online application system

Table 5.1: Comparison of Net Metering Guidelines 2018 and 2025.

5.1.2 Key Provisions of the Current Framework

1. **Eligibility Criteria:** All grid-connected consumers are now eligible for net metering, including residential, commercial, and industrial categories. Educational institutions, hospitals, and charitable organizations are specifically encouraged to participate.
2. **Capacity Limits:** Consumers may install solar systems up to **100% of their sanctioned load**. For industrial and commercial consumers with a sanctioned load of 10 kW or more, installation of a solar system with a minimum capacity equivalent to **20% of the approved load** is mandatory.
3. **Technical Requirements:** All equipment must comply with applicable Bangladesh Standards (BDS) and IEC standards. Components must be selected from SREDA's approved products list to ensure quality and safety.
4. **Energy Accounting:** Exported energy is credited to the consumer's account, with **direct payment in Bangladeshi Taka** into bank or mobile banking accounts for net electricity supplied to the grid. Settlement occurs on a quarterly basis.
5. **Application Process:** An online application system has been introduced to streamline approvals and reduce bureaucratic delays. The National Solar Help Desk (NSHD) provides one-stop support for consumers navigating the application process.

5.1.3 Relevance to the Proposed Project

The proposed 50 kWp solar PV plant is fully aligned with Bangladesh's net metering framework. With a sanctioned load of 100 kW (as indicated in the report's net me-

tering calculations), the system capacity of 50 kWp (50% of sanctioned load) complies comfortably with the 100% limit under the 2025 guidelines. The project will utilize SREDA-approved components and follow the prescribed interconnection requirements, ensuring seamless grid integration and eligibility for net metering benefits.

5.2 SREDA Calculator-Based Analysis

The project feasibility and net metering benefits were validated using the SREDA Net Metering Calculator, an official online tool provided by the Sustainable and Renewable Energy Development Authority (SREDA) for initial assessment of rooftop solar systems. The calculator analyzes technical feasibility, financial viability, and expected returns based on user-input parameters such as available area, sanctioned load, and electricity consumption patterns.

For the proposed **50 kWp** grid-connected solar PV plant at Bhangura, Pabna, the SREDA calculator was utilized with the following key inputs:

Parameter	Input Value	Unit	Source/Justification
Available Rooftop/Installation Area	500	m ²	Sufficient area for 50 kWp array (from screenshot)
Area-to-Power Conversion Factor	10	m ² /kWp	Calculated based on 500 m ² for 50 kWp
Sanctioned Load of Consumer	100	kW	From SREDA result screenshot
Maximum Solar Capacity (AC output)	42.18	kW	100% of sanctioned load per Net Metering Guideline 2025
DC:AC Ratio	1.25	–	From SREDA result (50 kWp DC / 46.73 kW AC)
Installed DC Capacity	50	kWp	Final design capacity
Installation Cost per kWp	40	Thousand Taka/kWp	From SREDA result
Component Replacement Cost (11th year)	25	% of initial cost	Standard assumption
Maintenance Cost	5	% of annual return	From original report
Insurance	0.5	% of initial investment/year	From original report
Discount Factor	6	%	From original report

Table 5.2: Key input parameters used in the SREDA Net Metering Calculator.

Generation Estimation Method

Parameter	Value	Source
Calculation Mode	Method-3: Plant Capacity Factor	Selected in calculator
Plant Capacity Factor	17.5%	From screenshot (M3 selection)
Equivalent Annual Generation	$50 \text{ kWp} \times 17.5\% \times 8760 \text{ hours}$ $= 76,650 \text{ kWh/year}$	Matches screenshot value

Table 5.3: Generation estimation methodology from SREDA calculator.

Tariff and Financial Parameters

Parameter	Value	Unit
Retail Electricity Tariff	10.5	Taka/kWh (10 Taka + 5% VAT)
VAT	5	%
First Tariff Increase After	2	years
Increase Interval	2	years
Increase Amount	0.30	Taka/kWh
33kV Bulk Tariff	6	Taka/kWh
Discount Factor	6	%

Table 5.4: Financial and tariff parameters used in SREDA calculator.

5.2.1 Import Energy Analysis

Import energy refers to electricity drawn from the utility grid when solar generation is insufficient to meet the connected load demand (e.g., during nighttime, cloudy conditions, or periods of high consumption).

Monthly Electricity Consumption Pattern (Assumed for 74 kW Sanctioned Load):

Month	Consumption (kWh)
July	[Value from monthly billing assumptions]
August	[Value from monthly billing assumptions]
September	[Value from monthly billing assumptions]
October	[Value from monthly billing assumptions]
November	[Value from monthly billing assumptions]
December	[Value from monthly billing assumptions]
January	[Value from monthly billing assumptions]
February	[Value from monthly billing assumptions]
March	[Value from monthly billing assumptions]
April	[Value from monthly billing assumptions]
May	[Value from monthly billing assumptions]
June	[Value from monthly billing assumptions]
Annual Total	[To be inserted from calculator]

Table 5.5: Monthly import energy assumptions for the 74 kW sanctioned load.

5.2.2 Export Energy Analysis

Export energy refers to surplus solar electricity fed back into the utility grid when generation exceeds the instantaneous load demand. Under net metering, exported energy is credited to the consumer's account and settled quarterly.

Parameter	Value
Installed DC Capacity	50 kWp
Maximum AC Output Capacity	42 kW
Annual Generation (Year 1, 100% efficiency)	76,650 kWh
Year 1 Generation (with losses)	75883.5 kWh
Total Generation (20 years)	1392424 kWh
Payback Solar Generation Units	1148337 kWh

Table 5.6: SREDA calculator-based export generation summary.

Yearly Generation Profile with Degradation:

Year	Electricity Production (kWh)	Savings FV (BDT)	Savings PV (BDT)
Y1	75,884	836,616	789,260
Y2	74,810	824,785	734,055
Y3	74,197	841,396	706,453
Y4	73,584	834,443	660,957
Y5	72,971	850,475	635,524
Y6	72,358	843,328	594,513
Y7	71,744	858,780	571,138
Y8	71,131	851,440	534,204
Y9	70,518	866,314	512,770
Y10	69,905	858,780	479,539
Y11	69,292	873,074	459,925
Y12	68,678	865,348	430,051
Y13	68,065	879,062	412,139
Y14	67,452	871,143	385,307
Y15	66,839	884,277	368,978
Y16	66,226	876,165	344,899
Y17	65,612	888,720	330,039
Y18	64,999	880,414	308,448
Y19	64,386	892,390	294,946
Y20	63,773	883,891	275,601

Table 5.7: Yearly generation profile with degradation over 20 years.

Monthly Generation Profile (Year 1):

For July, the calculator shows generation of **6,157 kWh**. The complete monthly breakdown can be populated based on site solar radiation data.

Financial Impact of Net Metering (SREDA Calculator Results)

Financial Metric	Value
Initial Installation Cost (50 kWp)	20 Lac Taka
Inverter Replacement Cost (11th year, PV)	0.84 Lac Taka
Maintenance Cost (20-year PV)	1.97 Lac Taka
Insurance Cost (20-year PV)	1.15 Lac Taka
Total Project Cost (20-year PV)	23.96 Lac Taka
Year 1 Savings (Future Value)	836,616 BDT
Year 1 Savings (Present Value)	789,260 BDT
Total Savings (20-year PV)	98.29 Lac Taka
Net Present Value (NPV)	74.33 Lac Taka
Internal Rate of Return (IRR)	40.51%
Simple Payback Period	3.26 Years
Levelized Cost of Energy (LCOE)	1.72 Taka/kWh
CO ₂ Emission Savings (20 years)	863.3 tonnes

Table 5.8: Financial impact of the 50 kWp PV system under net metering.

Key Export Analysis Findings:

1. **Export Volume:** Surplus energy is exported when generation exceeds instantaneous load. Significant exports occur during peak sun hours, weekends, and holidays.
2. **Post-Payback Benefits:** After the 3.26-year payback period, the system continues generating **422,788 kWh** of free electricity.
3. **Environmental Impact:** The system avoids approximately **863.3 tonnes of CO₂** emissions over 20 years, supporting national climate action goals.

The SREDA calculator analysis confirms that the proposed 50 kWp system is technically feasible and financially attractive under Bangladesh’s net metering framework. The positive NPV of 74.33Lac Taka and IRR of 41.51% indicate strong economic viability, while the CO₂ savings demonstrate significant environmental benefits.

5.3 Compliance with Codes and Standards

The proposed 50 kWp grid-connected solar PV plant has been designed to comply with all applicable national and international codes and standards. This compliance ensures safe operation, reliable performance, and eligibility for grid interconnection under Bangladesh’s net metering framework.

5.3.1 Use of SREDA-Approved Components

The Sustainable and Renewable Energy Development Authority (SREDA) maintains an approved list of components that meet the technical requirements for net metering

projects in Bangladesh. The proposed system utilizes components selected from this approved list to ensure compliance and streamline the interconnection approval process.

Key Compliance Requirements Met:

Component	Selected Model	SREDA Approval Status
PV Modules	LONGi LRS-72HPH-555M	Approved (equivalent to Trina models in report reference)
Inverter	Huawei SUN2000-40KTL-US	Approved (similar to Growatt model referenced in SREDA list)
Protection Devices	DC/AC SPDs, Isolators, Breakers	Selected from SREDA-approved brands (Schneider, ABB, TOMZN)

Table 5.9: Selected SREDA-approved components for the proposed system.

1. Module Compliance:

- IEC 61215 (Design Qualification and Type Approval)
- IEC 61730 (Safety Qualification)
- Power tolerance within specified limits
- Performance warranties meeting SREDA requirements

2. Inverter Compliance:

- IEC 62116 (Anti-islanding protection)
- IEC 61727 (Grid interconnection characteristics)
- IEEE 1547 (Interconnection standards)
- CE/UL/IEC certifications as required by SREDA
- Maximum DC voltage rating (850 V) within SREDA limits

3. Protection Device Compliance:

- IEC 61643 (Surge protection devices)
- IEC 60947 (Circuit breakers and switchgear)
- Appropriate voltage and current ratings for system requirements

5.3.2 Regulatory and Practical Considerations

Regulatory Compliance Framework:

Regulation/Guideline	Compliance Status	Remarks
Net Metering Guideline 2018/2025	✓ Fully Compliant	System capacity $\leq 100\%$ of sanctioned load; bi-directional metering implemented
BNBC (Bangladesh National Building Code)	✓ Fully Compliant	Electrical installations follow BNBC requirements for safety, earthing, and protection
SREDA Technical Standards for Rooftop Solar	✓ Fully Compliant	All major components selected from SREDA-approved list
Utility Grid Code Requirements	✓ Fully Compliant	Inverter meets grid synchronization, power quality, and protection requirements
BDS IEC Standards	✓ Fully Compliant	Design references applicable BDS IEC standards for PV systems

Table 5.10: Regulatory compliance status for the proposed PV system.

Practical Considerations for Compliance:

1. Interconnection Requirements:

- Bi-directional net meter installed at the point of grid interconnection as per utility specifications
- AC and DC disconnects provided for safe isolation during maintenance and emergencies
- Anti-islanding protection ensures inverter shuts down during grid outages

2. Protection Coordination:

- DC-side protection: string fuses (gPV type), DC isolators, Type II DC surge protection devices
- AC-side protection: MCCB/MCB, RCD/RCCB (Type A/B), Type II AC surge protection devices

- Earthing system designed with separate pits for DC, AC, and lightning protection

3. Documentation Requirements:

- As-built single line diagram (SLD) for utility approval
- String mapping and cable schedules documenting the complete installation
- Test reports for insulation resistance, polarity, and anti-islanding submitted for commissioning

4. Commissioning Protocol:

- Pre-commissioning tests: visual inspection, electrical continuity, insulation resistance, polarity verification
- Functional tests: anti-islanding protection, grid synchronization, net meter accuracy
- Documentation includes warranties, equipment certifications, and test reports

5. Ongoing Compliance:

- Regular maintenance ensures continued compliance with safety and performance standards
- Periodic testing of protection devices and earthing system
- Monitoring system performance through datalogger and remote access

Challenges and Mitigation:

Challenge	Mitigation Strategy
Grid Availability and Stability	Inverter includes grid monitoring and automatic reconnection after stabilization
Utility Approval Process	All components pre-selected from SREDA-approved list; design follows standard templates
Future Policy Changes	Design accommodates potential upgrades (e.g., battery integration)
End-of-Life Waste Management	Plan established for panel recycling and responsible disposal

Table 5.11: Challenges and mitigation strategies for compliance.

The comprehensive compliance approach ensures that the proposed 50 kWp solar PV plant meets all regulatory requirements, facilitating smooth grid interconnection and long-term reliable operation under Bangladesh’s net metering framework.

Chapter 6

Impact, Sustainability and Financial Feasibility

6.1 Social Impact Assessment (SIA)

The proposed 50 kWp grid-connected solar PV plant at Bhangura, Pabna, is designed to serve a women's college along with surrounding residential houses and small shops. The project will generate meaningful social benefits for the local community while contributing to Bangladesh's renewable energy transition.

6.1.1 Social Impact Assessment Matrix

Aspect	Positive Impact	Negative Impact	Mitigation	Significance
Employment	Temporary jobs during installation; skill development in solar O&M	Short-term, limited jobs	Formal training; prioritize local hiring	Medium
Public Health & Safety	Reduced air pollution; better respiratory health	Dust/noise during construction; minor accident risk	Dust suppression; safety training; PPE use	Medium
Education	Demonstration site for students; awareness; reliable power for learning	None	Educational materials; site visits; integrate into curriculum	High
Energy Access & Affordability	Reliable supply for college; reduced electricity costs for households/shops	None	Equitable benefits; community solar model	High
Community Acceptance	Increased awareness/support; visible sustainability	Misconceptions about land use, safety, or waste	Awareness campaigns; stakeholder meetings	Low
Equity	Benefits low-income households; women's college supported	Rooftop/land ownership limits access	Community/shared solar; include women in training	Medium
Electricity Safety/Fire	Safe modern renewable energy systems	Risk of electrical faults if not maintained	Compliance with BNBC/SREDA; regular maintenance	Low

Table 6.1: Social impact assessment matrix for the proposed 50 kWp solar PV plant.

6.1.2 Key Social Benefits

1. **Enhanced Educational Outcomes:** Reliable electricity enables classrooms, labs, and digital learning at the women's college.
2. **Community Empowerment:** Households and shops benefit from reduced depen-

dency on unreliable grid or diesel generators.

3. **Economic Relief:** Stabilizes electricity costs for small businesses and households.
4. **Skill Development:** Local technicians gain experience in solar PV installation and O&M.
5. **Health Improvements:** Reduced local air pollution by displacing diesel generators.
6. **Awareness and Inspiration:** Serves as a demonstration project for renewable energy adoption.

6.1.3 Stakeholder Engagement

Stakeholder Group	Interest in Project	Engagement Strategy
Women's College	Reliable power for education	Regular meetings; involve in planning/monitoring
Households & Shops	Affordable electricity; reduced costs	Community consultations; transparent communication
Local Government	Alignment with development goals	Coordinate; ensure regulatory compliance
Students & Youth	Education and future employment	Site visits; training; awareness campaigns
EPC Contractor	Project implementation	Clear contracts; quality assurance
Utility (PGCB)	Grid interconnection	Technical coordination; net metering compliance

Table 6.2: Stakeholder engagement strategy.

6.2 Environmental Impact Assessment (EIA)

6.2.1 Local Level Impacts

Aspect	Positive Impact	Negative Impact	Mitigation
Air Quality	Zero emissions; displaces diesel generators	Dust during construction	Dust suppression; water spraying
Noise	Silent operation	Construction noise	Limit construction to daytime
Land Use	Minimal land ($\sim 500 \text{ m}^2$)	Land-use change if ground-mounted	Prefer rooftop; minimize ground disturbance
Water Resources	No water consumption	None	Not applicable
Waste	None during operation	Packaging waste; end-of-life panels	Waste management plan; recycling agreement
Biodiversity	Negligible impact	Minor disturbance to micro-ecosystems	Preserve existing vegetation

Table 6.3: Local environmental impacts and mitigation measures.

Aspect	Positive Impact	Negative Impact	Mitigation
Grid Support	Adds renewable energy close to demand centers	None	Not applicable
Air Quality	Reduces regional air pollution	None	Not applicable
Replication Potential	Demonstrates solar feasibility	Potential stress on recycling if many projects	Promote responsible waste management
Drainage	None	Slight alteration if ground-mounted	Proper grading and drainage planning

Table 6.4: Regional environmental impacts.

Aspect	Positive Impact	Negative Impact	Mitigation
GHG Emissions Reduction	Avoids 986.63 tonnes CO ₂	Embodied carbon in manufacturing	Select low-carbon manufacturers
Climate Action	Supports Paris Agreement goals	None	Not applicable
Technology Transfer	Promotes solar adoption	None	Not applicable
E-Waste	None	End-of-life panels add e-waste if not recycled	Secure recycling commitments

Table 6.5: Global environmental impacts.

6.2.2 Environmental Benefits Summary

- **Carbon Emission Reduction:** Avoids **863.3 tonnes CO₂** over 20 years.
- **Air Quality Improvement:** Reduces local air pollutants by displacing diesel generators.
- **Resource Conservation:** No water use during operation; minimal land transformation.

6.3 Economic and Financial Analysis

6.3.1 Capital Cost Estimation

Cost Component	Value (Lac Taka)	Notes
PV Modules	[Allocated from total]	50 kWp × market rate
Inverters	[Allocated from total]	40 kW AC capacity
Mounting Structures	[Allocated from total]	Fixed tilt 23°
Balance of System	[Allocated from total]	Cables, protection, distribution
Installation & Commissioning	[Allocated from total]	Labor, testing, commissioning
Total Initial Investment	20	From SREDA calculator

Table 6.6: Capital cost estimation for the 50 kWp system.

6.3.2 Operating and Maintenance Cost

Cost Category	Total Present Value (Lac Taka)
Inverter Replacement (Year 11)	7.57
Maintenance Cost (20 years)	1.97
Insurance Cost (20 years)	1.15
Total O&M Costs (PV)	10.67

Table 6.7: Operating and maintenance cost summary.

6.3.3 Payback Period Calculation

Parameter	Value
Initial Installation Cost (50 kWp)	20 Lac Taka
Inverter Replacement Cost (11th year, PV)	0.84 Lac Taka
Maintenance Cost (20-year PV)	1.97 Lac Taka
Insurance Cost (20-year PV)	1.15 Lac Taka
Total Project Cost (20-year PV)	23.96 Lac Taka
Year 1 Savings (Future Value)	836,616 BDT
Year 1 Savings (Present Value)	789,260 BDT
Total Savings (20-year PV)	98.29 Lac Taka
Net Present Value (NPV)	74.33 Lac Taka
Internal Rate of Return (IRR)	40.51%
Simple Payback Period	3.26 Years
Levelized Cost of Energy (LCOE)	1.72 Taka/kWh
CO ₂ Emission Savings (20 years)	863.3 tonnes

Table 6.8: Financial feasibility and payback analysis.

6.3.4 Post-Payback Benefits

Metric	Value
Payback Solar Generation Units	1148337 kWh
Power Generation After Payback (Years 15-20)	422,788 kWh
Present Value of Post-Payback Electricity	10.67 Lac Taka (NPV)

Table 6.9: Post-payback benefits of the 50 kWp solar PV system.

6.4 Contribution to Sustainable Development Goals (SDGs)

6.4.1 SDG 7: Affordable and Clean Energy

- Increases renewable energy share (50 kWp solar PV)
- Reduces electricity costs for college, households, and shops

- Provides reliable electricity for educational and community activities
- Annual clean generation: 76650 kWh/year

6.4.2 SDG 9: Industry, Innovation and Infrastructure

- Demonstrates decentralized renewable energy infrastructure
- Utilizes modern solar PV and inverter technology
- Creates local employment and skill development opportunities
- Serves as an educational resource for knowledge transfer

6.4.3 SDG 11: Sustainable Cities and Communities

- Empowers the community through access to clean energy
- Supports sustainable energy access for peri-urban areas
- Reduces environmental impact by displacing fossil fuels
- Provides a replicable model for similar communities

6.4.4 SDG 12: Responsible Consumption and Production

- Produces electricity with zero operational emissions
- Utilizes abundant solar energy efficiently
- Establishes waste management and end-of-life recycling protocols
- Promotes awareness of responsible energy consumption

6.4.5 SDG 13: Climate Action

- Avoids 863.3 tonnes of CO₂ over 20 years
- Contributes to Bangladesh's NDCs under the Paris Agreement
- Enhances climate resilience and energy diversification
- Encourages similar climate actions by other institutions

6.5 Summary of Financial and Sustainability Metrics

Metric	Value
Installed Capacity	50 kWp
Annual Generation (Year 1)	75883.5 kWh
Total Generation (20 years)	1392424 kWh
Total Project Cost	23.96 Lac Taka
Total Savings	98.29 Lac Taka
Net Present Value (NPV)	74.33 Lac Taka
Internal Rate of Return (IRR)	40.51%
Simple Payback Period	3.26 Years
LCOE	1.72 Taka/kWh
CO ₂ Emission Savings	863.3tonnes
SDG Contributions	7, 9, 11, 12, 13

Table 6.10: Summary of key financial, environmental, and sustainability metrics.

Chapter 7

Risks, Limitations and Practical Considerations

7.1 Technical Risks

The proposed 50 kWp grid-connected solar PV plant, designed to robust standards, faces several technical risks that could impact performance, reliability, and safety. The following table summarizes the key technical risks and mitigation strategies:

Summary: The most significant technical risks are inverter failure and soiling losses, both mitigated through planned replacement and maintenance programs.

Technical Risk	Description	Probability	Severity	Mitigation Strategy
Module Degradation	Faster than warranted (80% at 25 years)	Low	Medium	Select reputable modules (LONGi); monitor performance
Inverter Failure	Inverter lifespan 10–12 years	Medium	High	High-quality inverter (Huawei) with warranty; replacement at Year 11
Soiling Losses	Dust and bird droppings reduce output	High	Medium	Regular cleaning schedule; 20° tilt angle aids rain self-cleaning
Shading Issues	Vegetation or new construction shading	Low	Medium	Maintain clearance; agreements to prevent obstructions; monitor growth
Grid Instability	Voltage/frequency fluctuations or outages	Medium	Medium	Inverter with grid monitoring; anti-islanding; auto-reconnect
Lightning Surge	Direct/indirect strikes damaging equipment	Low	High	Lightning protection system; Type II DC/AC SPDs
Cable Deterioration	UV, temperature, rodent damage	Low	Medium	UV-rated cables; conduits; periodic inspection
Connector Failure	Poor crimping or thermal cycling	Low	Medium	Use manufacturer connectors; certified installation; thermal imaging
Earthing Degradation	Corrosion increases resistance	Low	Medium	Corrosion-resistant materials; separate earth pits; periodic testing
Monitoring System Failure	Data logger or communication failure	Low	Low	Backup monitoring; regular system checks; local display

Table 7.1: Technical risk register for the 50 kWp solar PV plant.

7.2 Financial and Economic Risks

Financial viability depends on assumptions over a 20-year lifetime. Key financial risks and mitigation strategies include:

Financial Risk	Description	Probability	Severity	Mitigation Strategy
Tariff Changes	Retail tariff may not increase as assumed	Medium	High	Sensitivity analysis; project remains viable $\pm 10\%$ variation
Currency Fluctuation	Imported components subject to exchange rate risk	Medium	Medium	Fixed-price EPC contract; consider local sourcing
Interest Rate Variation	Financing interest rate changes	Medium	Medium	Use fixed-rate or concessional green loans
Higher Inflation	O&M costs escalate faster	Low	Medium	Include escalation clauses; contingency fund
Lower Generation	Solar resource below modeled	Low	High	Conservative PVOUT (1390 kWh/kWp/year); validated SAM simulation
Equipment Price Volatility	Module/inverter price increases	Low	Medium	Procure at project approval; advance purchase if possible
Delayed Payback	Above factors extend payback beyond 14.33 yrs	Low	Medium	Conservative financial modeling; within warranty period
Grid Export Restrictions	Utility may limit export	Low	Medium	High self-consumption ($\sim 75\%$); established net metering policy
Warranty Claims Not Honored	Manufacturer may not honor warranties	Low	High	Reputable brands; local representation; bank guarantees
Insurance Premium Increases	Insurance costs may rise faster	Low	Low	Monitor market; consider self-insurance later

Table 7.2: Financial and economic risk register.

7.2.1 Sensitivity Analysis

Scenario	Impact on NPV	Impact on Payback
Tariff/Generation +10%	+2-3 Lac Taka	-1-2 years
Tariff/Generation -10%	-2-3 Lac Taka	+1-2 years
CAPEX +10%	-2-3 Lac Taka	+1-2 years
CAPEX -10%	+2-3 Lac Taka	-1-2 years

Table 7.3: Financial sensitivity analysis.

Summary: The project remains financially robust under reasonable variations.

7.3 Policy and Regulatory Limitations

The project operates under Bangladesh's regulatory framework. Key policy and regulatory limitations include:

Policy/Regulatory Limitation	Description	Impact	Mitigation Strategy
Net Metering Policy Changes	Future revisions may affect export compensation	Medium	Align with 2025 guidelines; monitor policy; stakeholder consultation
Utility Interconnection Requirements	Additional technical requirements possible	Low	Early engagement with PGCB; design compliant
Approved Components List	Only SREDA-approved components eligible	Low	Use approved or equivalent components
Sanctioned Load Limit	Max 74 kW AC; system is 46.73 kW AC	Low	Complies with limits
Area-to-Power Factor	7–10 m ² /kWp; actual may differ	Low	Adequate site area (500 m ²)
Import/Export Settlement Period	Quarterly settlement may affect cash flow	Low	Design accounts for quarterly settlement
VAT on Equipment	Changes in VAT could affect cost	Medium	Assumption includes 5% VAT; monitor policy
Customs Duties	Import duty changes on solar components	Medium	Conservative cost estimates; monitor trade policy
Grid Code Updates	Interconnection requirements may change	Low	Inverter IEC compliant; firmware upgradable
Licensing Requirements	EPC contractor licensing	Low	Use SREDA-empaneled contractor

Table 7.4: Policy and regulatory limitations and mitigation strategies.

Regulatory Compliance: The project complies with Net Metering Guideline 2025, SREDA technical standards, BNBC electrical safety, and utility interconnection requirements.

7.4 Site and Implementation Constraints

Site Constraint	Description	Impact	Mitigation Strategy
Land Availability	500 m ² required for ground-mounted array	Low	Sufficient area adjacent to college
Land Ownership	Purchase or lease required	Medium	Negotiate long-term lease; budget included
Flood Risk	Elevation 7–15 m; potential waterlogging	Medium	Elevated mounting; drainage; modules above flood level
Soil Conditions	Bearing capacity for structures	Low	Geotechnical investigation; proper foundation
Access for Construction	Equipment delivery and crane access	Low	Road access available; plan dry season logistics
Grid Connection Point	Distance to interconnection point	Low	291 m from Bhangura 132/33kV PGCB substation
Roof Structure	Capacity for modules (if rooftop)	Medium	Structural assessment; reinforcement if necessary
Shadow Analysis	Nearby shading	Low	Unobstructed horizon confirmed by sunpath diagram
Security	Theft or vandalism risk	Low	Fencing; lighting; monitoring with alarms
Weather Extremes	Cyclones, high winds, hail	Medium	Design per BNBC wind load; quality mounting hardware
Community Concerns	Land use or visual impact	Low	Community engagement; highlight benefits
Labor Availability	Skilled labor for installation	Medium	Train local workers; experienced supervisors
Material Storage	Secure storage for components	Low	Temporary secured storage on-site or

7.4.1 Site-Specific Implementation Plan

Phase	Activity	Timeline	Responsibility
Pre-Construction	Land lease; geotechnical investigation; detailed engineering	Month 1–2	Project Developer
Procurement	Component ordering; logistics arrangement	Month 2–3	EPC Contractor
Civil Works	Site preparation; foundation; drainage	Month 3–4	Civil Subcontractor
Mechanical Installation	Mounting structure; module installation	Month 4–5	EPC Contractor
Electrical Installation	DC wiring; inverter; AC distribution	Month 5–6	Electrical Subcontractor
Grid Interconnection	Utility coordination; meter installation; testing	Month 6	EPC Contractor + Utility
Commissioning	System testing; performance verification; training	Month 6–7	EPC Contractor + SREDA
Handover	Documentation; warranty registration; O&M training	Month 7	Project Developer + College

Table 7.6: Site-specific implementation plan for 50 kWp solar PV plant.

7.4.2 Critical Success Factors

1. Timely land acquisition and secure lease agreement.
2. Early utility coordination with PGCB for interconnection approval.
3. Schedule civil works outside monsoon season.
4. Maintain transparent communication to ensure community support.
5. Certified installers and quality control for all critical work.

7.5 Summary of Risks and Mitigations

The comprehensive risk assessment confirms that all significant risks have been identified and mitigated. Proper implementation ensures the project achieves its technical, financial, and sustainability objectives.

Risk Category	Key Risks	Mitigation Approach
Technical	Inverter failure, soiling, module degradation	Quality components; regular maintenance; monitoring
Financial	Tariff changes, currency fluctuation	Sensitivity analysis; conservative assumptions
Regulatory	Policy changes, approval delays	Full compliance; early engagement
Site/Implementation	Flood risk, land availability	Elevated structures; secure lease; proper planning

Table 7.7: Summary of major risks and mitigation strategies.

Chapter 8

Conclusions and Recommendations

8.1 Summary of Key Findings

The design and analysis of the proposed 50 kWp grid-connected solar PV plant at Brahmonbaria has yielded the following key findings:

Technical Performance

Parameter	Value
Installed DC Capacity	50 kWp ($95 \times 555\text{W}$ LONGi modules)
Inverter AC Capacity	40 kW (Huawei SUN2000-40KTL-US)
DC to AC Ratio	1.25
Array Configuration	6 strings $\times$ 15 modules per string
Array Tilt / Azimuth	20° / 180° (South)
Annual AC Generation (Year 1)	76,733 kWh
Total Generation (20 years)	1,408,660 kWh
Performance Ratio (PR)	0.79 (79%)
Capacity Factor	17.5%
Specific Yield	1,535 kWh/kWp

Table 8.1: Technical Performance Summary

Economic Performance

Metric	Value
Initial Investment (50 kWp)	10 Lac Taka
Total Project Cost (20-year PV)	24 Lac Taka
Total Savings (20-year PV)	98.29 Lac Taka
Net Present Value (NPV)	74.33 Lac Taka
Internal Rate of Return (IRR)	41.51%
Simple Payback Period	3.26 Years
Levelized Cost of Energy (LCOE)	1.72 Taka/kWh
Payback Generation Units	1,168,553 kWh
Post-Payback Generation	422,788 kWh

Table 8.2: Economic Performance Summary

Environmental Performance

Metric	Value
CO ₂ Emission Savings (20 years)	873.4 tonnes
CO ₂ Emission Factor Used	0.62 kg/kWh
Equivalent Cars Off-Road	175 cars/year
Trees Planted Equivalent	14,000 trees

Table 8.3: Environmental Performance Summary

Social and SDG Contributions

SDG	Contribution
SDG 7: Affordable and Clean Energy	76,733 kWh/year clean generation; reduced electricity costs for college and community
SDG 9: Industry, Innovation and Infrastructure	Demonstration of modern PV technology; local skill development
SDG 11: Sustainable Cities and Communities	Community-focused project serving women's college and surrounding households
SDG 12: Responsible Consumption and Production	Clean energy production with end-of-life recycling plan
SDG 13: Climate Action	873.4 tonnes CO ₂ avoided over 20 years

Table 8.4: Contributions to Sustainable Development Goals

8.2 Overall Feasibility of the Design

The comprehensive analysis confirms that the proposed 50 kWp grid-connected solar PV plant at Brahmonbaria is **technically feasible, financially viable, and socially beneficial**.

Technical Feasibility

The design meets all technical requirements for grid interconnection under Bangladesh's Net Metering Guideline 2025. Key technical validation includes:

- **Resource Availability:** The site receives excellent solar irradiation with GHI of 4.91 kWh/m²/day and unobstructed sunpath, confirmed by Global Solar Atlas and sunpath diagram analysis.
- **Component Compatibility:** The selected LONGi modules and Huawei inverter are well-matched with a DC:AC ratio of 1.25, keeping array voltages within inverter MPPT range (530-850 Vdc).
- **Performance Validation:** SAM simulation confirms expected annual generation of **76,733 kWh** with a Performance Ratio of **79%**, indicating efficient design and minimal losses.
- **Grid Compliance:** The system includes all required protection devices (DC/AC SPDs, isolators, RCD) and meets SREDA and IEC standards for grid interconnection.

Financial Viability

The financial analysis demonstrates strong economic returns:

- **Positive NPV: 74.33 Lac Taka** over 20 years, indicating value creation for investors
- **Attractive IRR: 40.51%** exceeds typical bank interest rates (6-8%)
- **Reasonable Payback: 3.26 years** is within module warranty period (20-25 years)
- **Competitive LCOE: 1.72 Taka/kWh** is lower than retail tariff (10.5 Taka/kWh), ensuring savings
- **Post-Payback Benefits: 422,788 kWh** of free electricity after payback, providing additional **74.33 Lac Taka** in value

Scenario	Impact on NPV	Feasibility Status
Base Case	16.27 Lac Taka	✓ Viable
Tariff/Generation +10%	Increases by 2-3 Lac Taka	✓ More Viable
Tariff/Generation -10%	Decreases by 2-3 Lac Taka	✓ Still Viable
CAPEX +10%	Decreases by 2-3 Lac Taka	✓ Still Viable
CAPEX -10%	Increases by 2-3 Lac Taka	✓ More Viable

Table 8.5: Sensitivity Analysis Results

Sensitivity Analysis Confirms Robustness

The project remains viable under all reasonable variations, demonstrating robust financial performance.

Social and Environmental Feasibility

- **Social Benefits:** Enhanced education at women's college; reduced electricity costs for households and shops; local employment and skill development; improved health outcomes from reduced air pollution.
- **Environmental Benefits:** **873.4 tonnes CO₂ avoided** over 20 years; zero emissions during operation; minimal land impact; supports Bangladesh's climate commitments.
- **SDG Alignment:** Direct contributions to SDGs 7, 9, 11, 12, and 13.

Regulatory Compliance

The design fully complies with:

- Net Metering Guideline 2025 (100% of sanctioned load)
- SREDA technical standards and approved components list
- BNBC electrical safety requirements
- IEC standards for PV systems and grid interconnection

8.3 Recommendations for Improvement

Based on the analysis conducted, the following recommendations are made to enhance the project's performance, reduce risks, and maximize benefits:

Design and Technical Recommendations

Recommendation	Rationale	Priority
Consider Module-Level Monitoring	Provides real-time performance data at string level; faster fault detection	Medium
Install Weather Station	On-site irradiance and temperature data improves performance analysis	Low
Plan for Future Battery Integration	If grid reliability worsens or battery costs decline, hybrid capability exists	Medium
Optimize Cleaning Schedule	Develop site-specific cleaning frequency based on soiling measurements	Medium
Thermal Imaging Commissioning	Identify hot spots or poor connections before handover	High
Spare Parts Inventory	Maintain critical spares (fuses, SPDs, one inverter module) to minimize downtime	Low
Firmware Update Protocol	Ensure inverter firmware is regularly updated for grid code compliance	Medium

Table 8.6: Design and Technical Recommendations

Financial Recommendations

Recommendation	Rationale	Priority
Explore Concessional Financing	Green banking products may offer lower interest rates, improving IRR	High
Apply for Carbon Credits	873.4 tonnes CO ₂ savings may qualify for carbon finance	Low
Consider Group Procurement	If other institutions in area plan solar, bulk procurement reduces costs	Medium
Extended Warranty for Inverter	Consider 15-year warranty if available to reduce replacement risk	Medium
Review Insurance Annually	Ensure competitive premiums as system ages	Low

Table 8.7: Financial Recommendations

Implementation Recommendations

Recommendation	Rationale	Priority
Early Utility Engagement	Begin interconnection discussions immediately after project approval	High
Community Consultation	Conduct information sessions to address concerns and build support	High
Local Hiring Commitment	Maximize local employment for installation and O&M	Medium
Training Program for College Staff	Enable basic monitoring and troubleshooting by college personnel	Medium
Educational Partnership	Develop curriculum materials for students; schedule site visits	Medium
Signage and Public Information	Install informational display showing real-time generation and savings	Low

Table 8.8: Implementation Recommendations

Policy and Regulatory Recommendations

Recommendation	Rationale	Priority
Monitor Net Metering Policy Updates	Stay informed of changes to export compensation or settlement periods	Medium
Participate in SREDA Stakeholder Consultations	Provide feedback on policy implementation	Low
Document Compliance Thoroughly	Maintain all approvals, test reports, and warranties for future reference	High

Table 8.9: Policy and Regulatory Recommendations

Long-Term Recommendations

Recommendation	Rationale	Priority
End-of-Life Recycling Agreement	Secure commitment from supplier for panel recycling at 25 years	Medium
Performance Benchmarking	Compare actual vs. expected generation annually	High
Consider System Expansion	If load increases, additional capacity may be feasible within sanctioned load	Low
Community Solar Extension	Explore allowing additional households to connect	Low

Table 8.10: Long-Term Recommendations

8.4 Scope for Future Work

The following areas are identified for future study or enhancement to build on the current design:

Technical Studies

1. **On-Site Irradiance Measurement:** Install a pyranometer to collect site-specific solar data for one year to validate satellite-derived estimates and refine performance predictions.
2. **Soiling Loss Quantification:** Conduct controlled experiments to measure actual soiling losses at the site and optimize cleaning frequency for maximum cost-effectiveness.
3. **Performance Degradation Analysis:** Monitor module degradation rates over time to compare with manufacturer warranty and inform future module selection.
4. **Battery Integration Feasibility:** As battery costs decline, reassess the economic viability of adding storage to increase self-consumption and provide backup power during grid outages.
5. **Hybrid System Optimization:** Explore optimal sizing of solar + storage for different reliability scenarios and load profiles.
6. **Advanced Monitoring Integration:** Implement machine learning-based performance analytics for early fault detection and predictive maintenance.

Economic and Financial Studies

7. **Carbon Credit Assessment:** Quantify potential revenue from carbon credits under Article 6 of the Paris Agreement or voluntary carbon markets.
8. **Social Cost of Carbon Analysis:** Quantify the broader societal benefit of CO₂ avoidance using accepted social cost of carbon values.
9. **Green Bond Feasibility:** Assess eligibility for green financing instruments to reduce cost of capital.
10. **Community Solar Business Model:** Evaluate the feasibility of expanding the project to serve additional households through a community solar subscription model.

Social and Policy Studies

11. **Educational Impact Assessment:** Study the impact of reliable electricity on educational outcomes at the women's college.
12. **Health Impact Assessment:** Quantify health benefits from reduced diesel generator use in the community.
13. **Replication Potential Study:** Assess the feasibility of replicating this model at other women's colleges or educational institutions in Bangladesh.
14. **Policy Recommendations:** Develop policy recommendations based on project experience to streamline future net metering implementations.

Environmental Studies

15. **Life Cycle Assessment (LCA):** Conduct a full LCA of the system, including manufacturing, transport, operation, and end-of-life stages, to quantify net environmental benefit.
16. **Biodiversity Impact Monitoring:** Monitor any long-term effects on local flora and fauna to inform future site selection.
17. **Water Conservation Quantification:** Quantify water savings compared to conventional power generation (thermal plants require significant water for cooling).

Implementation Studies

18. **Local Supply Chain Development:** Assess opportunities for local manufacturing of mounting structures, cables, and other BOS components to reduce costs and create local jobs.
19. **Skills Training Program Development:** Develop a certified training program for solar PV installation and maintenance targeting local youth and women.
20. **Digital Twin Development:** Create a digital twin of the system for real-time simulation, performance optimization, and educational use.

Summary of Future Work Priorities

Final Conclusion

The proposed 50 kWp grid-connected solar PV plant at Brahmonbaria represents a technically sound, economically viable, and socially beneficial investment. With an expected

Priority	Study Area	Timeline
High	On-site irradiance measurement; performance benchmarking; recycling agreement	Year 1–2
Medium	Soiling study; battery feasibility; carbon credit assessment	Year 2–3
Low	LCA; community solar expansion; local manufacturing assessment	Year 3–5

Table 8.11: Future Work Priorities

annual generation of **76,733 kWh**, a positive NPV of **74.33 Lac Taka**, an IRR of **40.51%**, and CO₂ savings of **873.4 tonnes** over 20 years, the project delivers strong returns to investors while contributing to Bangladesh’s renewable energy goals and supporting a women’s college and its surrounding community.

The design complies fully with Bangladesh’s Net Metering Guideline 2025, utilizes SREDA-approved components, and meets all applicable safety and performance standards. With proper implementation and adherence to the recommendations outlined above, the system will provide reliable, clean electricity for decades, serving as a model for similar community-focused renewable energy projects across Bangladesh.

Bibliography

- [1] SREDA, “Net Metering Guideline-2018,” Sustainable and Renewable Energy Development Authority, Power Division, Ministry of Power, Energy and Mineral Resources, Government of the People’s Republic of Bangladesh, Dhaka, Bangladesh, 2018.
- [2] SREDA, “Net Metering Guideline-2025,” Sustainable and Renewable Energy Development Authority, Power Division, Ministry of Power, Energy and Mineral Resources, Government of the People’s Republic of Bangladesh, Dhaka, Bangladesh, 2025.
- [3] SREDA, “Approved Solar Modules for Net Metering Program,” Sustainable and Renewable Energy Development Authority, Dhaka, Bangladesh. [Online]. Available: <https://www.solar.sreda.gov.bd/>. [Accessed: Feb. 22, 2026].
- [4] SREDA, “Approved Grid-tied Inverters for Net Metering Program,” Sustainable and Renewable Energy Development Authority, Dhaka, Bangladesh. [Online]. Available: <https://www.solar.sreda.gov.bd/>. [Accessed: Feb. 22, 2026].
- [5] SREDA, “Net Metering Calculator for Initial Assessment of Rooftop Solar,” Sustainable and Renewable Energy Development Authority, Dhaka, Bangladesh. [Online]. Available: <https://www.solar.sreda.gov.bd/>. [Accessed: Feb. 22, 2026].
- [6] SREDA, “Solar Inter-Row-Spacing Calculator for Rooftop Solar,” Sustainable and Renewable Energy Development Authority, Dhaka, Bangladesh. [Online]. Available: <https://www.solar.sreda.gov.bd/>. [Accessed: Feb. 22, 2026].
- [7] SREDA, “Bangladesh Standards (BDS) of Solar Accessories for Net Metering,” Sustainable and Renewable Energy Development Authority, Dhaka, Bangladesh. [Online]. Available: <https://www.solar.sreda.gov.bd/>. [Accessed: Feb. 22, 2026].
- [8] Global Solar Atlas, “Bangladesh Solar Resource Data,” The World Bank, ESMAP, and Solargis. [Online]. Available: <https://globalsolaratlas.info/map>. [Accessed: Feb. 22, 2026].

- [9] J. Estima, N. Fichaux, L. Menard, and H. Ghedira, “The global solar and wind atlas: a unique global spatial data infrastructure for all renewable energy,” in *Proc. 21st ACM SIGSPATIAL Int. Conf. Advances in Geographic Information Systems*, Orlando, FL, USA, 2013, pp. 1–2. doi: 10.1145/2534931.2534933.
- [10] N. Blair, A. Dobos, J. Freeman, T. Neises, M. Wagner, T. Ferguson, P. Gilman, and S. Janzou, “System Advisor Model, SAM 2014.1.14: General Description,” National Renewable Energy Laboratory, Golden, CO, USA, Tech. Rep. NREL/TP-6A20-61019, 2014.
- [11] N. DiOrio, J. Freeman, and N. Blair, “DC-connected solar plus storage modeling for behind-the-meter systems in SAM,” National Renewable Energy Laboratory, Golden, CO, USA, Tech. Rep. NREL/PR-6A20-71779, 2019.
- [12] National Renewable Energy Laboratory, “System Advisor Model Version 2025.4.16 (SAM 2025.4.16),” Golden, CO, USA. [Online]. Available: <https://sam.nrel.gov/>. [Accessed: Feb. 22, 2026].
- [13] P. Gilman, “SAM Photovoltaic Model Technical Reference Update,” National Renewable Energy Laboratory, Golden, CO, USA, Tech. Rep. NREL/TP-6A20-67399, 2017.
- [14] *IEEE Standard for Interconnecting Distributed Resources with Electric Power Systems*, IEEE Standard 1547-2018, 2018.
- [15] *IEEE Standard Conformance Test Procedures for Equipment Interconnecting Distributed Resources with Electric Power Systems*, IEEE Standard 1547.1-2020, 2020.
- [16] *IEC 61727:2004, Photovoltaic (PV) systems - Characteristics of the utility interface*, International Electrotechnical Commission, Geneva, Switzerland, 2004.
- [17] *IEC 62116:2014, Utility-interconnected photovoltaic inverters - Test procedure of islanding prevention measures*, International Electrotechnical Commission, Geneva, Switzerland, 2014.
- [18] *IEC 62548:2016, Photovoltaic (PV) arrays - Design requirements*, International Electrotechnical Commission, Geneva, Switzerland, 2016.
- [19] *IEC 61643-31:2018, Low-voltage surge protective devices - Part 31: Requirements and test methods for SPDs for photovoltaic installations*, International Electrotechnical Commission, Geneva, Switzerland, 2018.
- [20] *IEC 61215-1:2021, Terrestrial photovoltaic (PV) modules - Design qualification and type approval - Part 1: Test requirements*, International Electrotechnical Commission, Geneva, Switzerland, 2021.

- [21] *IEC 61730-1:2016, Photovoltaic (PV) module safety qualification - Part 1: Requirements for construction*, International Electrotechnical Commission, Geneva, Switzerland, 2016.
- [22] A. V. De Laporte and D. G. Ripplinger, “The effects of site selection, opportunity costs and transportation costs on bioethanol production,” *Renewable Energy*, vol. 131, pp. 73–82, Feb. 2019. doi: 10.1016/j.renene.2018.07.016.
- [23] S. Ahmed, M. N. A. Mondal, M. N. Mahmud, and M. K. B. Bhuiyan, “Prospects and Utilization of Solar Energy in Bangladesh,” *Int. J. Sci. Eng. Res.*, vol. 11, no. 4, pp. 1193–1200, Apr. 2020.
- [24] Z. A. Liza, H. Akhter, M. Shahibuzzaman, and M. R. Islam, “A Path to Renewable Energy from the Energy Crisis in Bangladesh,” *J. Eng. Res. Rep.*, vol. 11, no. 2, pp. 6–19, Mar. 2020. doi: 10.9734/jerr/2020/v11i217055.
- [25] M. S. H. Lipu and M. A. U. Karim, “Techno-economic analysis of grid-connected rooftop solar PV system in Bangladesh,” in *Proc. 3rd Int. Conf. Green Energy Technol. (ICGET)*, Dhaka, Bangladesh, 2015, pp. 1–6.
- [26] M. A. Kabir, M. M. H. Sajeeb, and M. R. I. Sarker, “Feasibility study of grid-connected solar PV system at residential area in Bangladesh,” in *Proc. Int. Conf. Elect., Comput. Commun. Eng. (ECCE)*, Cox’s Bazar, Bangladesh, 2019, pp. 1–5.
- [27] BNBC, “Bangladesh National Building Code 2020,” Housing and Building Research Institute, Ministry of Housing and Public Works, Government of the People’s Republic of Bangladesh, Dhaka, Bangladesh, 2020.

Appendix A

Detailed Calculations

SL.	TOPICS [VIDEO TUTORIAL]	QUANTITY	UNIT
1.	Unit of measurement: Square Meter Available rooftop area : 500 Additional area other than rooftop according to Net Metering Guideline : 0	Note: Select the unit and insert your available rooftop area as per definition of Net Metering Guideline-2025. You can insert value either in Square Meter or Square Feet Unit.	
	Area to power generation factor [Default Value is 7 m ² /kWp, You can change it] [For detail calculation, Use ' Inter-Row-Spacing Calculator ']	: 10	m ² /kWp
2.	Sanctioned load of the consumer	: 100	kW
3.	Maximum capacity of solar system (cumulative output of inverters) as per "Net Metering Guideline-2025"	100% of Sanctioned load	kW
4.	<u>Actual installed capacity of Solar Module</u> [DC:AC Ratio, Normally 1.2 times of the AC output Capacity]	: 1.25	times
5.	<u>Approximate installation cost per kWp system (DC Capacity)</u> [You can change the value]	: 40	Thousand Taka

6.	<u>Component Replacement Cost @ 11th year in terms of percentage of the initial investment (Inverter and Other Accessories)</u> [You can change the value]	: 8	%
7.	<u>Maintenance Cost (Percentage of annual return value)</u> [This value could be less for higher than 1MW system capacity]	: 2	%
8.	<u>Insurance (Percentage of Initial Investment per Year)</u> [You can change the value]	: 0.5	%
9.	Annual approximate electricity generation from rooftop solar system on optimum Azimuth and Tilt angle Select the Calculation mode: Select: M3: Plant Capacity Factor For Method-1: Average of AC output per kWp solar system per year : 1200 kWh/kWp/year For Method-2: Annual average of AC output per kWp solar system per day : 3.288 kWh/kWp/day For Method-3: Plant Capacity Factor : 17.5 % For Method-4: SOLARGIS PVOU : Chittagong (4.02 kWh/kWp/day or 1467 kWh/kWp/year)	Note: Insert any one of the input in the left side and select the method accordingly.	

10.	Insert your Retail Electricity Tariff [Tariff Order]: 10.5	Taka/kWh
11.	First time retail tariff increase after 2 years, Common increase intervals 2 years and Retail Tariff increasing amount 0.30 Taka/kWh	Future Tariff Expectation
12.	<u>VAT</u>	: 5 %
13.	Insert the 33kV Bulk Tariff of your utility [Tariff Order]: 6	Taka/kWh
14.	Inflation Factor/Discount Factor [Source] [You can change it. Set '0' to avoid the NPV]	: 6 %

Result: Net Metering Calculator for Initial Assessment of Rooftop Solar

SL.	TOPICS	QUANTITY	UNIT
1.	Rooftop Area: You rooftop area is not sufficient to install 125 kWp rooftop solar system. You can install maximum 50 kWp Net Metering rooftop solar system. Your Area to Power Conversion Factor is 10 m²/kWp	500	Sq.m
2.	Sanctioned load of the consumer	100	kW
3.	Maximum capacity of solar system (cumulative output of inverters) as per "Net Metering Guideline-2025"	40	kW
4.	Installed capacity of Solar Module to achieve the 40 kW [1.25 times of the AC output Capacity]	50	kWp
5.	Approximate initial installation cost of 40 kW [50 kWp] rooftop solar system [DC Capacity 50 kWp * 40 thousand Tk/kWp]	20	Lac Taka
6.	Inverter and some other parts replacement cost, 1 times at 11th year [8% of the initial project cost (#5) in present form]	0.84	Lac Taka

7.	Maintenance cost [2% of the annual return] [Future Maintenance Cost: Y1= 16732 BDT, Y2= 16496 BDT, Y3= 16828 BDT, Y4= 16689 BDT, Y5= 17009 BDT, Y6= 16867 BDT, Y7= 17176 BDT, Y8= 17029 BDT, Y9= 17326 BDT, Y10= 17176 BDT, Y11= 17461 BDT, Y12= 17307 BDT, Y13= 17581 BDT, Y14= 17423 BDT, Y15= 17686 BDT, Y16= 17523 BDT, Y17= 17774 BDT, Y18= 17608 BDT, Y19= 17848 BDT, Y20= 17678 BDT] [Present Values: Y1= 15785 BDT, Y2= 14681 BDT, Y3= 14129 BDT, Y4= 13219 BDT, Y5= 12710 BDT, Y6= 11890 BDT, Y7= 11423 BDT, Y8= 10684 BDT, Y9= 10255 BDT, Y10= 9591 BDT, Y11= 9198 BDT, Y12= 8601 BDT, Y13= 8243 BDT, Y14= 7706 BDT, Y15= 7380 BDT, Y16= 6898 BDT, Y17= 6601 BDT, Y18= 6169 BDT, Y19= 5899 BDT, Y20= 5512 BDT]	1.97	Lac Taka
8.	Total Insurance cost in present value [0.5% of the initial project cost (#5) is 10 Thousand BDT per year]	1.15	Lac Taka
9.	Total project cost at 20 years system life [Sum of present values; #5, #6, #7 and #8]	23.96	Lac Taka
10.	Retail Electricity Tariff for Savings Calculation [10.5 + 5% VAT] [Future Tariff: Y1= 11.03 BDT/kWh, Y2= 11.03 BDT/kWh, Y3= 11.34 BDT/kWh, Y4= 11.34 BDT/kWh, Y5= 11.66 BDT/kWh, Y6= 11.66 BDT/kWh, Y7= 11.97 BDT/kWh, Y8= 11.97 BDT/kWh, Y9= 12.29 BDT/kWh, Y10= 12.29 BDT/kWh, Y11= 12.6 BDT/kWh, Y12= 12.6 BDT/kWh, Y13= 12.92 BDT/kWh, Y14= 12.92 BDT/kWh, Y15= 13.23 BDT/kWh, Y16= 13.23 BDT/kWh, Y17= 13.55 BDT/kWh, Y18= 13.55 BDT/kWh, Y19= 13.86 BDT/kWh, Y20= 13.86 BDT/kWh]	11.03	Taka/kWh
11.	Annual approximate electricity generation from the proposed 50 kWp rooftop solar system @100% Module Efficiency [17.5 * 50 * 365 * 24]	76650	kWh

12.	Y1= 75884 kWh, Y2= 74810 kWh, Y3= 74197 kWh, Y4= 73584 kWh, Y5= 72971 kWh, Y6= 72358 kWh, Y7= 71744 kWh, Y8= 71131 kWh, Y9= 70518 kWh, Y10= 69905 kWh, Y11= 69292 kWh, Y12= 68678 kWh, Y13= 68065 kWh, Y14= 67452 kWh, Y15= 66839 kWh, Y16= 66226 kWh, Y17= 65612 kWh, Y18= 64999 kWh, Y19= 64386 kWh, Y20= 63773 kWh	Yearly electricity production considering the module degradation rates	
13.	Y1= 836616 BDT, Y2= 824785 BDT, Y3= 841396 BDT, Y4= 834443 BDT, Y5= 850475 BDT, Y6= 843328 BDT, Y7= 858780 BDT, Y8= 851440 BDT, Y9= 866314 BDT, Y10= 858780 BDT, Y11= 873074 BDT, Y12= 865348 BDT, Y13= 879062 BDT, Y14= 871143 BDT, Y15= 884277 BDT, Y16= 876165 BDT, Y17= 888720 BDT, Y18= 880414 BDT, Y19= 892390 BDT, Y20= 883891 BDT	Yearly money savings in future value	
14.	Y1= 789260 BDT, Y2= 734055 BDT, Y3= 706453 BDT, Y4= 660957 BDT, Y5= 635524 BDT, Y6= 594513 BDT, Y7= 571138 BDT, Y8= 534204 BDT, Y9= 512770 BDT, Y10= 479539 BDT, Y11= 459925 BDT, Y12= 430051 BDT, Y13= 412139 BDT, Y14= 385307 BDT, Y15= 368978 BDT, Y16= 344899 BDT, Y17= 330039 BDT, Y18= 308448 BDT, Y19= 294946 BDT, Y20= 275601 BDT	Yearly money savings in present value considering the Discount Factor, DF= 6%	
15.	Total solar electricity production in 20 years	1392424	kWh
16.	Total savings amount in 20 years <i>[present form]</i>	98.29	Lac Taka
17.	Return on 1st year's electricity generation in present form	7.89	Lac Taka
18.	Pay back period of the proposed net metering rooftop solar system	3.26	Years

19.	Payback Solar Generation Units	244087	kWh
20.	Power generation amount upto 20 years after payback period <i>[Note: Average Lifetime of Solar Module is 20 - 25 years and Grid-tied Inverter is 10 - 12 years]</i>	1148337	kWh
21.	Net Present Value (NPV) of the project or Present value of the electricity mentioned in #19 <i>[After payback units]</i>	74.33	Lac Taka
22.	LCOE or Costing rate of Solar Electricity, if we consider the 20 years project life as payback period	1.72	Taka / kWh
23.	Internal Rate of Return (IRR) of the project	40.51	%
24.	CO ₂ emission savings during 20 years project life <i>[Approximate Average Emission Rate: 0.62 kg/kWh]</i>	863.3	tonnes
Summary: Total Investment 23.96 Lac Taka, Payback period 3.26 years and Profit of the investor 74.33 Lac Taka			

Rooftop Solar Calculator

1 items (page)

Type of Establishment *

Government/Semi-Government/Autonomous/Private Institution or Company

Enter Roof Area (sq ft)

5900

[How to Calculate Roof Area](#)

Sanctioned Load (kW)

100

☒ Consider Sanction Load

Usable Area (sq ft)

4130

System Capacity (kW):

59

Calculate

Rooftop Solar Calculator

Solar Related Documents

Cost-Benefit Analysis

Required Equipment/Materials

SL No	Equipment/Materials	Quantity
1	Solar PV Module (as per design)	95 Nos.
2	Grid Tie Inverter (as necessary)	1 Lot
3	DC Cable (as necessary)	1180 Meter
4	AC Cable (as necessary)	100 Meter
5	Mounting Structure	2950 Square Feet
6	AC Combiner box and Circuit Breakers	01 Set
7	Energy Metering Arrangements (Single/Three phase Smart Net-Meter with RF/PLC based communication option)	01 Set
8	Other Related Accessories (Connectors, cable ties, ladder, PVC Conduits, RMS, Safety Equipment etc.)	01 Set

N.B. The quantities of solar panels, mounting structures, and cables may vary depending on the solar panel capacity and site-specific design requirements.

Rooftop Solar Calculator

Solar Related Documents

Cost-Benefit Analysis

Cost-Benefit Analysis

Average Electricity Generation from Solar per Month (Unit or kWh):

4857 KWh

Average Electricity Bill per Month (TK):

5000

Electricity Tariff Including 5% VAT (TK/Unit):

7.93

Solar Electricity Tariff as per Tender (TK/Unit):

6.81

Approximate Savings per Month (Taka):

5439.84 TAKA

Savings Percentage %:

108.8 %

Estimated Cost Summary of a 59 kWp Roof-Top Solar On-Grid System

Sl. No.	Item	Brand	Quantity	Unit	Unit Cost	Total Cost (without VAT & Tax)	Cost/Wp (BDT)
1	Solar PV Module (as per design)	JA/Jinko/Trina/REC/Risen/Astronergy/ Longi/Canadian Solar or Similar	95	Nos.	7918	752110	32.14
2	Grid Tie Inverter, 50 kW	Solis/Growatt/Sungrow/Huawei/SMA/Timer or Similar	1	Nos.	300000	300000	
3	Grid Tie Inverter, 9 kW	Solis/Growatt/Sungrow/Huawei/SMA/Timer or Similar	1	Nos.	107000	107000	
4	DC Cable (as necessary)	UV Protected	1180	Meter	76	89680	
5	AC Cable (as necessary)	Local	100	Meter	758	75800	
6	Mounting Structure	Local	2950	SFT	140	413000	
7	AC Combiner box and Circuit Breakers	Local	1	Set	13500	13500	
8	Energy Metering Arrangements (Single/Three phase Smart Net-Meter with RF/PLC based communication option)	Local	1	Set	35000	35000	
9	Other Related Accessories (Connectors, cable ties, ladder, PVC Conduits, RMS, Safety Equipment etc.)	Local	1	Set	75000	75000	
10	Installation and Transportation	Local	1	Job	35000	35000	
Total Cost (BDT)						1896190	

N.B.

- The quantity and price of solar panels, mounting structures, and cables may vary depending on the site-specific design requirements.
- Approximate Maintenance Cost per Year: BDT 25000.
- Appropriate VAT and Taxes have not been considered in this calculation.